To Co- Is Human

Designing Technologies That Center
Human Connection, Co-creativity, and Calm in the Era of AI

by

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Submitted to the Program in Media Arts and Sciences, School of Architecture and Planning, in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

at the
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ABSTRACT:
In an era where Artificial Intelligence (AI) systems are increasingly framed as our "companions" and "co-creators," this dissertation reclaims "co-" as a fundamental marker of shared human experience—using it as a foundation to reimagine and build technologies that consciously center interhuman connection and co-creativity. Central to this work, we've developed CoCo (coco.build)—a general-purpose, real-time co-creative learning platform that empowers young people to engage in a wide variety of safe, shared creative experiences with their peers, spanning creative computing, AI education, digital art, writing, and more. Through the platform, we showcase how digital environments can move beyond isolated modes of learning and creating to support multiple ways of being creative together with others—introducing a new paradigm for real-time digital collaboration. We further illuminate how CoCo has been envisioned as a "self-less" social platform that de-emphasizes comparison-based, self-centric metrics (profiles, likes, followers) prevalent in most online systems for youth. We anchor these interconnected ideas in a unifying theme of "Being. Creative. Together."—reflecting timeless values that have become especially timely in an era when AI tools can further accentuate individualized digital experiences for young people. We supplement the broader design, technical, practical, and pedagogical contributions of this work by sharing insights and feedback from pilots with over 2,000 young people and educators across diverse settings. Ultimately, we see this dissertation as both a contribution and a call—to preserve the human essence of co-, to distinguish it from the useful, powerful, but instrumental AI interactions, and to shape digital environments that nurture young people's capacity to co-imagine, co-create, co-learn, co-exist, and co-evolve—with and through one another.

Note: This work has been co-developed with Manuj Dhariwal. See coco.build/thesis for suggested citation and updates on this work.
Thesis supervisor: Mitch Resnick
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Prelude

To co- is human.

To cO-, AI?

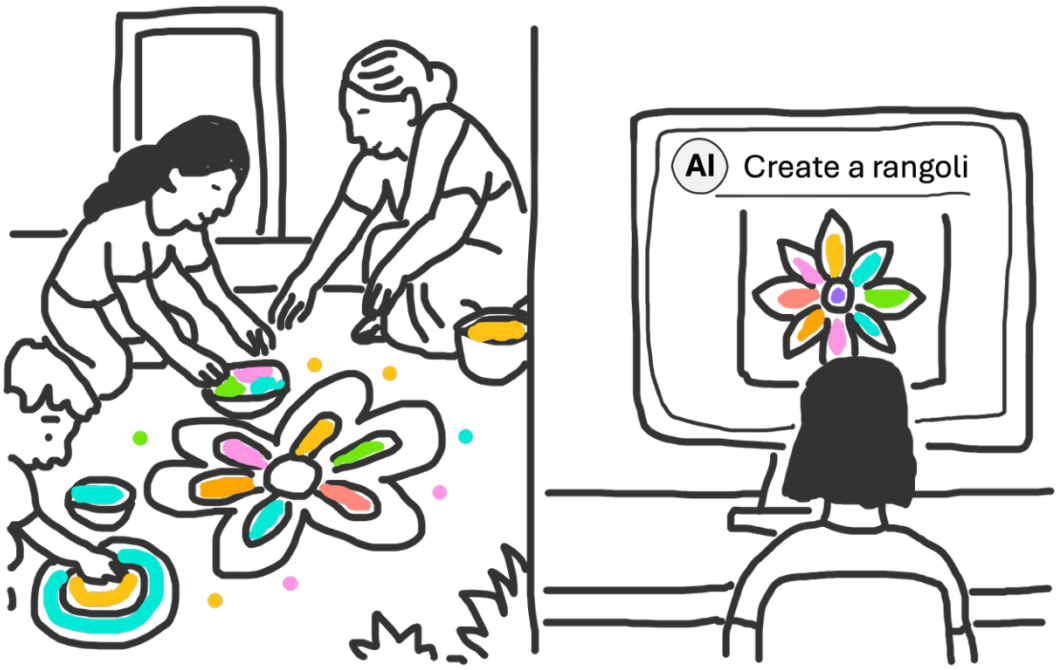

Fig 1: A sketch showing a family co-creating a traditional Indian rangoli—a floor design made with colors (left), and a young person working independently at a computer using an AI tool to generate a similar colorful pattern (right).

"**E**veryone wants a friend to just talk to in real life," reflected thirteen-year-old Sabrina. "You can never get that sense of community and support from AI. You can't really talk to AI about friendship problems… or well, I guess you won't have any if your only friend is AI," she added with a chuckle.

We shared a laugh when she said that—but the irony is, Sabrina's words aren't far from where things are headed. It's May 2025. Mark Zuckerberg has recently announced his ambition to combat loneliness by giving everyone AI "companions"—another step toward blurring the line between human and artificial relationships.

Every day, AI tools and chatbots are being pitched as our "co-workers," "co-creators," or in other terms that, not long ago, unequivocally described human relationships. The more comfortable we get in adopting this language, the more we risk skewing our relationships—not just with machines, but with one another.

How can we update our vocabulary to reinforce and protect this important boundary?

We'd like to propose a small thought experiment: What if we preserve the prefix "co-" for our timeless relationships with other living beings, and introduce a new prefix, "c0-" (with a zero), to better describe our instrumental relationships with AI tools and chatbots? The numeral 0 would serve as a crucial reminder of the underlying mathematical and statistical nature of machine intelligence. While undoubtedly powerful, useful, astonishing even—it is ultimately reducible to 0s and 1s—to bits, not biology.

It is a subtle reframing, but one that we urgently need as AI systems become interwoven with our creative, social, and emotional worlds.

Think of the difference between co-creating with another person versus c0-creating with an AI tool, or having a c0-worker on your machine versus a co-worker you can share a coffee with. While the prefix "c0-" is just one possibility, and how to pronounce it is an open question, it creates a meaningful distinction in writing.

For instance, consider how this might play out for AI companies: Microsoft Copilot would be rebranded as C0pilot—emphasizing it is an AI assistant and not a replacement for human collaboration. And AI companies would stop presenting their chatbots as companions, instead calling them c0mpanions—reminding users not to mistake them for human, even as they get more convincing in mimicking human capabilities and behaviors.

The point is not necessarily to choose one over the other—but to clearly distinguish one from the other.

Why does this distinction matter?

The prefix co- captures a broad range of our uniquely shared experiences—of co-imagining, co-creating, and co-existing. In assigning this deeply human-oriented prefix to AI, we aren't just unduly elevating the tools, we're also undermining the meaning and value of human connection.

Amplifying objects into subjects and reducing subjects into objects are both distortions of reality. Consider how many of society's deepest failures have stemmed from a common root cause: the harmful tendencies to dehumanize and objectify people. Now with AI we run the risk of rapidly accelerating in the opposite direction: humanizing AI and treating AI-powered tools, chatbots, and other objects as if they have thoughts, feelings, or personalities.

At their core, AI models are trained by optimizing what's called an "objective function"—a mathematical way of defining a specific goal. For instance, the builders of a so-called companion bot may choose to set its objective as "maximize user engagement." Any attribution of subjectivity to such a bot conceals its fundamental nature as a tool built to be a 'means to an end'; in this case, the *end* is to keep the user chatting with the bot. An April 2025 Common Sense Media report on social AI companions highlights growing concerns around excessive chatbot use—particularly among young people—including unhealthy attachments, reduced critical thinking, and even psychological harm. These risks are likely to intensify as AI systems become more engaging, persuasive, and addictive.

A small shift in language—reclaiming back "co-" for human experiences and normalizing a new prefix for AI—can be a step toward helping young people meet this reality with more clarity. In doing so, we will also pass this distinction to the AI models who train on our language. Because if we stop calling a bot "companion," it will learn to *not* call itself one.

A call for co-creating a new vocabulary

Maybe we need a new word altogether—or several new words—to capture the variety of ways we're engaging with AI. Given that this is a reality we all must navigate together, it makes sense to collectively shape the language of this new era. As a step in

this direction, we invite you to share suggestions for new words and any thoughts in this open form: coco.build/czero. Together, we might co-imagine and co-create a vocabulary that sticks.

Every act of naming is an act of worldbuilding. Let's gift young people a world where they continue to have *friendship problems*; a world where they can leverage the power of c0-creating with AI, while also experiencing the joy of co-creating together with friends; a world where to "co-" is *undoubtedly* human[1].

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[1] Note: A version of this prelude appeared as an op-ed in The Boston Globe, May 2025. https://www.bostonglobe.com/2025/05/21/opinion/manuj-shruti-dhariwal-mash-c0-ai-interactions/

!!

Before diving in further, we'd like to address

the **elephant** in this dissertation.

Here it is :)

A note about our joint dissertations and shared authorship

Hello! We are Shruti and Manuj. You can call us "**MASH**"—a shared author identity we have coined (from MAnuj and SHruti) to reflect and reinforce the co-creative nature of our work. At the time of co-writing this dissertation, we have been married for over 12 years—being creative together, in life and work, for a long time. A few days ago, we jointly defended our dissertation work at MIT Media Lab. It was the *first ever joint PhD defense* in the history of the lab, and possibly, MIT.

The ideas, technologies, and research presented in this dissertation have been conceived and developed together. But as per academic guidelines, each of us has submitted a separate dissertation. While the content and structure across our dissertations is largely overlapping, we chose to adopt distinct titles: *To Co- is Human* (Shruti Dhariwal) and *Being. Creative. Together.* (Manuj Dhariwal). This choice reflects our shared process while letting each of us to bring forth unique framing lenses and complementary perspectives.

At the core, however, this dissertation is an expression of our *shared* interests, values, and visions. It is not possible for us to separate our voices and contributions into neat buckets. A shared identity lets us represent our work more truthfully and authentically. We recognize that this approach runs counter to the dominant academic norms—where there is a strong emphasis on individual authorship and clearly delineated credit, even when the work is fundamentally collaborative. We are deeply

grateful to our advisor, Mitch Resnick, our dissertation committee, and the Media Lab, for supporting and trusting this creative collaboration—against the norms.

Co-creating this work has not let our individual perspectives be diminished in any way. On the contrary, we like to think that while we are *two* individuals, when we work together, we actually become *three*. There is Shruti, Manuj, and this third entity that comes into being, MASH, which represents the mutual space *between* us—a space where we continue to develop ideas we wouldn't, or couldn't, arrive at alone.

And just as a MASH cannot be turned back into potatoes, this work, too, cannot be separated back into parts. In that spirit, we hope the dissertation serves as an embodiment of the very ideas it seeks to spark.

**

Please visit coco.build/thesis for updates on this work and suggested citation that reflects equal authorship. Thank you.

To co- is human

To imagine is to co-imagine
To create is to co-create
To clutter is to co-clutter
To exist is to co-exist
To evolve is to co-evolve

We are not separate 'I's,
drifting through space and time

To human is to co-
To co- is human

**

The **5Co's** structure of this dissertation

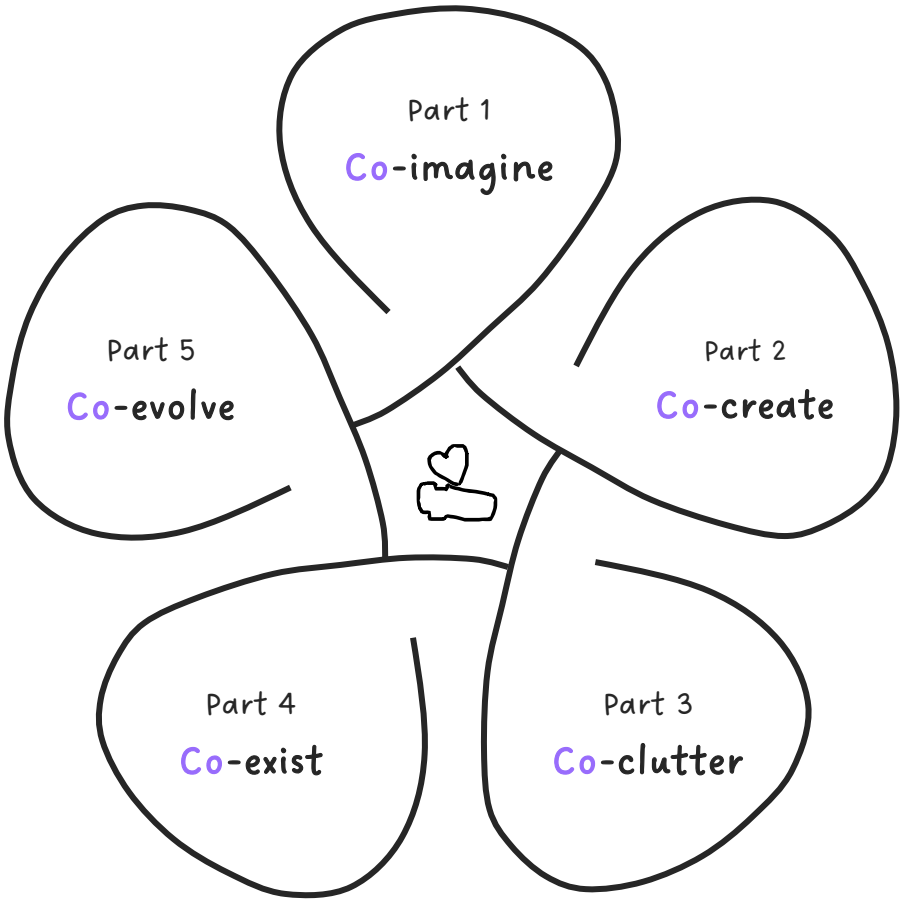

Fig 2: The 5Co's structure of this dissertation

This dissertation is rooted in the prefix "co-". It has been shaped through a lived co-experience. Like any shared journey, this document is not strictly linear or cleanly delineated. Some chapters go into detail, some are sparks, some connect to the ones that come before, some share new perspectives. They are distinct yet deeply interconnected. We invite you to explore these interconnections in your own ways.

Contents

Abstract 2
Prelude: **To co- is human. To c0-, AI?** 4
The *elephant* in this dissertation | A note about shared authorship 8
A poem on **To co-** 11

Part 1. **Co-imagine** | roots and values | 15
Ch 1. Self-expression, software, systems, and spirituality 16
Ch 2. A tale of children, mindfulness, and multiplayer shooting games 21
Ch 3. Two questions diverged in the wood 24
Ch 4. An ecology of values: "Being. Creative. Together." 25

Part 2. **Co-create** | technologies and possibilities | 27
Ch 5. A glimpse into CoCo: coco.build 28
Ch 6. Building a new "co-creative" paradigm for digital collaboration 31
6.1 Designing for togetherness over collaboration 33
6.2 Knobs over switches: Multiple ways of being creative together 34
6.2.1 Shared Presence 35
6.2.2 Playful Inspiration 36
6.2.3 Fluid Recomposition 37
6.2.4 Creative Interaction 38
6.2.5 Coordinated Collaboration 39
Ch 7. New paradigm, new possibilities: Insights from pilots with children 41
Ch 8. Extending creative tools into co-creative environments 49
Ch 9. Designing decentralized communities: Privacy, safety, cultural context 51

Part 3. **Co-clutter** | a necessary interlude | 53
Ch 10. Creating the conditions for "collision" 54

Part 4. **Co-exist** | shifts and perspectives | 56
Ch 11 From future to present 57
Ch 12. Constructing *things*, Constructing *context* 58
Ch 13. Head, hands, heart, and *hüm* 61
Ch 14. From self-inflating to self-*less* social platforms 63
Ch 15. Passion, dispassion, and compassion 68

Part 5. **Co-evolve** | contributions and directions | 69
Ch 16. The 5Co's of Co-creative Learning 70
Ch 17. From Monoculture to CoCo-culture 74
Ch 18. Bringing CoCo to communities around the world 78
Ch 19 Extending CoCo | Little Language Models (LLMs): Empowering children to be future AI modelers 80
Ch 20. Looking ahead: Centering the *timeless* in timely 92

Afterword(s) 94
References 95
Acknowledgments 98

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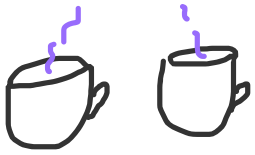
Part 1

Co-imagine

| roots and values |



A lot of our *co-imagining* happens through conversations over our morning chai (Indian tea). Unlike the typical notion of morning meetings, where the agenda for the day is decided, this shared routine is essentially an agenda-free zone for us. One of us might share a half-formed thought, an interesting observation, or a random what-if. The other *listens*—not always, but very often :). There is a back and forth that follows. It matters that there is mutual trust and (long) shared context. It leads to lightness and spaciousness, both in our heads and in between us. When *we* are totally *present*, but our egos are not, co-imagining happens without effort. We disagree and diverge freely, and agree and appreciate easily. It's sort of like denoising an image. Starting with a foggy impulse, and with each exchange, layering in more clarity, depth, and richness. Not alone, but *through* each other's imaginations.

How can we support young people to imagine *through* each other's imaginations?

In this section, we share a glimpse into the space of ideas, contexts, and experiences that served as the backdrop for us to co-imagine new connections and possibilities.

Ch 1.	Self-expression, software, systems, and spirituality	16
Ch 2.	A tale of children, mindfulness, and multiplayer shooting games	21
Ch 3.	Two questions diverged in the wood	24
Ch 4.	An ecology of values: "Being. Creative. Together."	25

Chapter 1

Self-expression, software, systems, and spirituality

…the greatest poet drew from no other source than
the one each of us has within his own being.

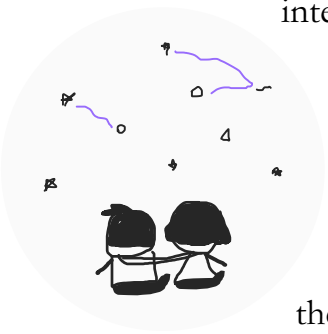
—

HERMANN HESSE

The four words in the title above may not seem to have an obvious connection. Yet, around six years ago, these themes increasingly started to surface together in our shared experiences, conversations, and imaginations. Looking back, this wasn't a coincidence.

The ideas that inspire us, the values we resonate with, the people and contexts we are drawn to—they are all reflections of who we are. They speak to our core self, while also reshaping the self over time in ways that are often hard to describe.

In this chapter, we share the initial space of ideas and contexts that served as the backdrop of this work. We highlight how they began to intersect with our shared interests and values and eventually led us to draw new connections and ask new questions. It's not a consolidated literature review. The connections will emerge organically throughout the dissertation—mirroring our own journey of developing this work. But we hope these glimpses can help spark others, just as they sparked us, to explore the space in new ways and draw their own connections and constellations over time.



For the past several years, the Lifelong Kindergarten group at MIT Media Lab has been our wonder-*full* shared context for discovering, imagining, creating, and playing with new ideas together.

We first formally encountered the term 'constructionism' coined by Seymour Papert and the ecosystem of rich educational ideas surrounding it as part of this context. We instinctively and deeply resonated with the group's core principles and approach of designing learning experiences for children rooted in 4Ps—Projects, Passion, Peers, and Play [1], [2], [3]. The Scratch online platform for children [4], that we got to be closely involved with and contribute to as part of the group, served as a wonderful playground for us to see these ideas in action through the wide range of children's creative and social learning experiences [5]. These foundational ideas around how children learn by creating; and how we can design expressive tools and environments that empower children to learn by creating [6], especially in ways that are personally meaningful to them [7]—while being so inspiring also felt very intuitive to us. It was because we were experiencing these ourselves every day. They became naturally internalized in our thinking and we could see them being reflected in all our work and explorations.

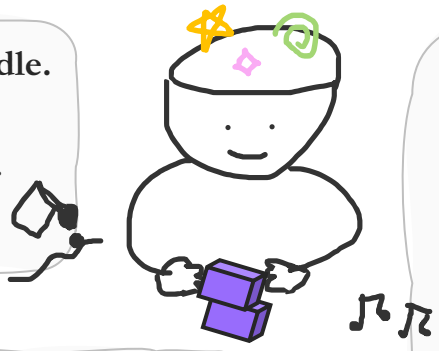
Being part of this ecosystem, we continued to be inspired by the rich heritage of powerful educational ideas throughout history around centering children's voice, creative expression, and their self-identities in the seminal works by Papert [7], Malaguzzi [8], Freire [9], Montessori [10], Dewey [11], Holt [12], and others.

Inspiring ideas on centering children's interests and creative self-expression

The child has
a hundred languages
a hundred hands
a hundred thoughts
a hundred ways of thinking
of playing, of speaking...
- **Loris Malaguzzi**

…construction that takes place "in the head" often happens especially felicitously when it is supported by **construction of a more public sort "in the world"-** a sandcastle or a cake, a Lego house or a corporation, a computer program, a poem, or a theory of the universe.
- **Seymour Papert**

Curiosity is never idle.
Doing is learning.
Knowledge is action.
- **John Holt**



Children are not listeners.
They are first and foremost **interested in moving**, communicating, exploring the world, **constructing and expressing themselves** artistically.
- **John Dewey**

An **investment in interest** always pays off with the best knowledge.
- **Mitch Resnick**

The more students work at storing the deposits entrusted to them, the less they develop the **critical consciousness** which would result from their intervention in the world as transformers of that world.
- **Paulo Freire**

The absorption of the environment is an intellectual activity. It is a psychic necessity that **the child explores the environment**; it satisfies his spirit.
- **Maria Montessori**

Making connections with our other interests…

While we were continuing to explore these ideas, we also found ourselves naturally gravitating toward our other shared life interests—in meditation, mindfulness, awareness, self.

Beyond our practice and personal experience, we also explored the themes around self and interconnectedness across a wide range of books and works in—philosophy, physics, psychology, systems, and spirituality [13], [14], [15], [16].

We deeply cared about these ideas and values and they increasingly became dominant in our everyday morning conversations over breakfast.

To understand life is to understand ourselves, and that is both the beginning and the end of education.
- J. Krishnamurty

The unit of survival is **organism plus environment**
- **G. Bateson**

Who am I?
- **Sri Ramana Maharishi**

At the outset, these two spaces seemed to be pointing in opposite directions.

One emphasizes the idea of **centering** every child's individual self-expression and self-identity as a creator. The other stresses on the importance of **de-centering** the individual self, deconstructing the notion of self, and seeing beyond the self. One highlights how children are always interested in moving and exploring the world around them, the other talks of stillness and exploring the inner world.

In his wonderful Artful Design book [17], Ge Wang shares a prompt that perfectly captures the essence of these two spaces of ideas that we were thinking about.

Things that **move** you.
Things that make you **still**.

Is there a way to combine both these things for children? We wondered.

Out of curiosity, as a small exploration around this time, we also designed a new **Mindfulness + Scratch** activity for an upcoming Scratch Day event at MIT. Prior to the workshop, several people in the group had expressed their genuine apprehension about whether this theme would be relevant for such a high energy event with kids actively exploring, moving, creating, and playing in the space.

This was Spring 2019, pre-pandemic era. At the time, the themes of cultivating mindfulness and awareness didn't feature directly in our surrounding conversations about children, creative learning, and technology. But the best part about being in a supportive research context is the freedom to explore your interests even if they may seem fuzzy initially. The workshop turned out to be great and attracted kids across ages along with their parents. They all really appreciated and enjoyed the experience. We felt there was something here.

We intuitively knew that these two *seemingly disparate* worlds were synergistic in essence, and we continued to wonder about new ways and mediums that could reflect these interconnections and synergy.

Chapter 2

A tale of children, mindfulness, and multiplayer shooting games

Fig 3: Image generated from ChatGPT for illustrative purposes

Around this time in Spring 2019, we had volunteered as group leads at a day-long mindfulness retreat for children near Boston. There were two separate incidents from the day that eventually sparked the beginnings of our ideas and work.

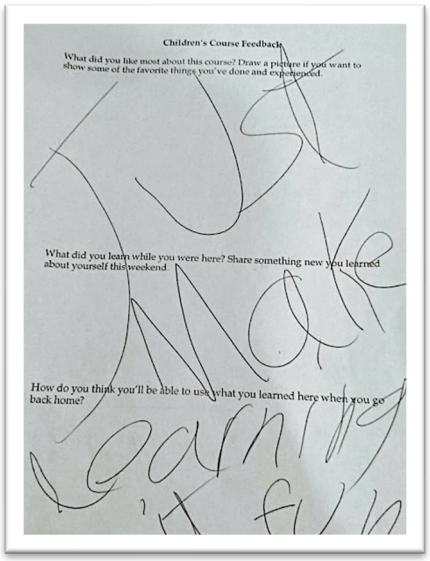
The **first** happened early in the day during a much-awaited play break after a stillness-based session. A group of 10- to 12-year-olds started talking passionately about some of the multiplayer shooting games they had been playing online. While the topic was not particularly 'appropriate' for a mindfulness retreat :), we decided it was best to not judge the conversation and let it flow.

We casually intervened and asked them, “How much time do you spend playing these games over the weekend?” to which a 12-year-old immediately responded…

“I play for 3–4 hours every day, what else is there to do!”

We talked some more. He shared the he was the only child and that his parents work till late and so in the evenings he jumped into these immersive multiplayer games. It was a way for him to avoid feeling lonely.

The **second** incident happened at the end of the day. The organizers had asked all the kids to fill a feedback form about their experience at the retreat.

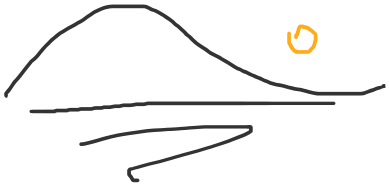


A form shared by one of the kids jumped out to us. Scribbled over the entirety of the page in big letters was a 5-word sentence that said,

“Just make learning it fun.”

The **it** here referred to the practice of mindfulness that kids were introduced to through the stillness-based sessions during the day.

These two incidents and the context in which they happened stayed with us.



Our continuing conversations after the retreat revolved around two themes…

1. Connection and Construction

As we thought about the first incident, we initially found ourselves contrasting the often passive, addictive, and close-ended nature of many online games—with the more open-ended and expressive digital making environments, such as Scratch, that are designed to let children explore a wide range of creative possibilities such as art, games, music, stories, or animations, based on their own interests.

However, in reflecting further on our own co-creative process and the comment made by the 12-year-old, we recognized that there was a big **missing piece of *real-time connection* with peers in the existing online creative learning tools and environments for children**.

2. Re-imagining social systems

The second incident prompted us to reflect on how the ideas that get reinforced by most of the existing social systems young people engage in (both online and offline) are often at odds with the deeper ideas of interconnectedness and nonjudgment that lie at the core of mindfulness. The notions of likes, followers, ranks, and scores are rooted in **individual-centric and comparison-based paradigms that can often lead to many young people feeling anxious.**

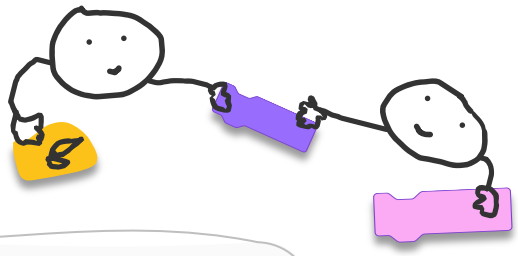
While there is value in teaching mindfulness practices to children as an approach toward helping them feel more calm and centered, we wondered what it might mean to reimagine social systems as calm playgrounds for children to encounter and explore these ideas in ways that are connected with their interests, and can help them question the dominant paradigms they are exposed to. As it turned out, this approach resonated with many parents who reached out to us later…

> "I read about your work and it speaks to me, for Rahil alas would have no patience for any mindfulness retreat or indeed any meditative experience, and envelops himself in the stimuli of Fortnite and Roblox, with his friends on mic. I often have wondered how this stimulus-driven cooperativity of his might be harnessed for learning rather than only gameplay."

Chapter 3

Two questions diverged in a wood

We drafted two guiding design-based research questions that became central to our work and explorations (even though at the time they didn't seem directly connected).

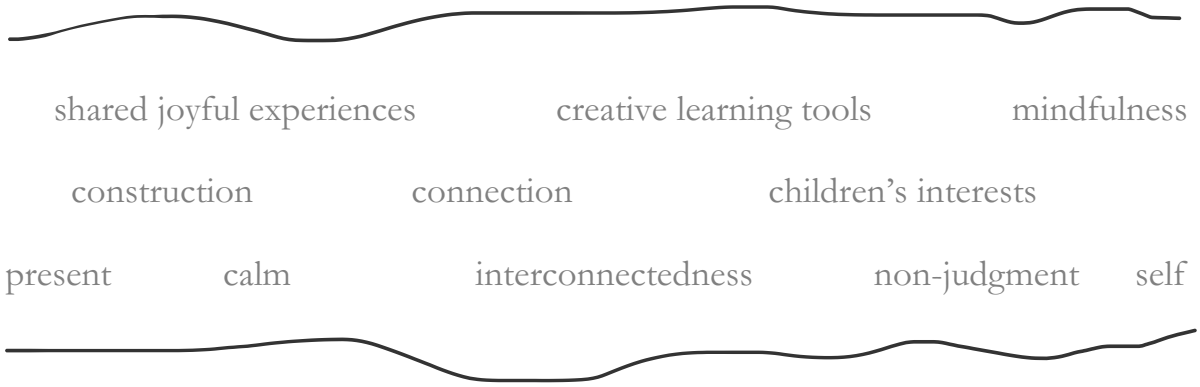


1. How can we design online creative environments for young people that **center authentic connection** with others—helping them experience the power and joy of co-creating and co-learning together?

2. How can we design online social systems for young people that **de-center the individual self—**helping them be fully engaged in the present, free of anxiety and judgment, in a shared experience with others?

Chapter 4

An ecology of values:
BEING. CREATIVE. TOGETHER.



Over time, all of our *seemingly disparate* interests, ideas, questions, and explorations morphed together into a synergistic framework of values that forms the foundation of this work. We coined the term "Being. Creative. Together." as it authentically and succinctly captures the space of intertwined ideas that we deeply care about and hope to illuminate through this work. We consciously choose to put periods next to each word in the phrase to bring awareness to the three core values of *Beingness*, *Creativeness*, and *Togetherness* that form the foundation of our work and motivations.

Beingness represents the **state** of *being*—being alive, aware, and fully engaged in the present moment. *Creativeness* is the **state** of *being creative*—experiencing joy *in* the process of creating and exploring without worrying about the outcome. And finally, *Togetherness* is the **state** of *being together*—feeling authentically connected with others through a shared experience.

Each of these values, along with their intersections, is an end-in-itself and not a means-to-an-end [17]—that is, they are not valuable for the sake of something else, but for their own sake. These are values that do not change with the changing times or evolving technologies, and thus are universal across contexts.

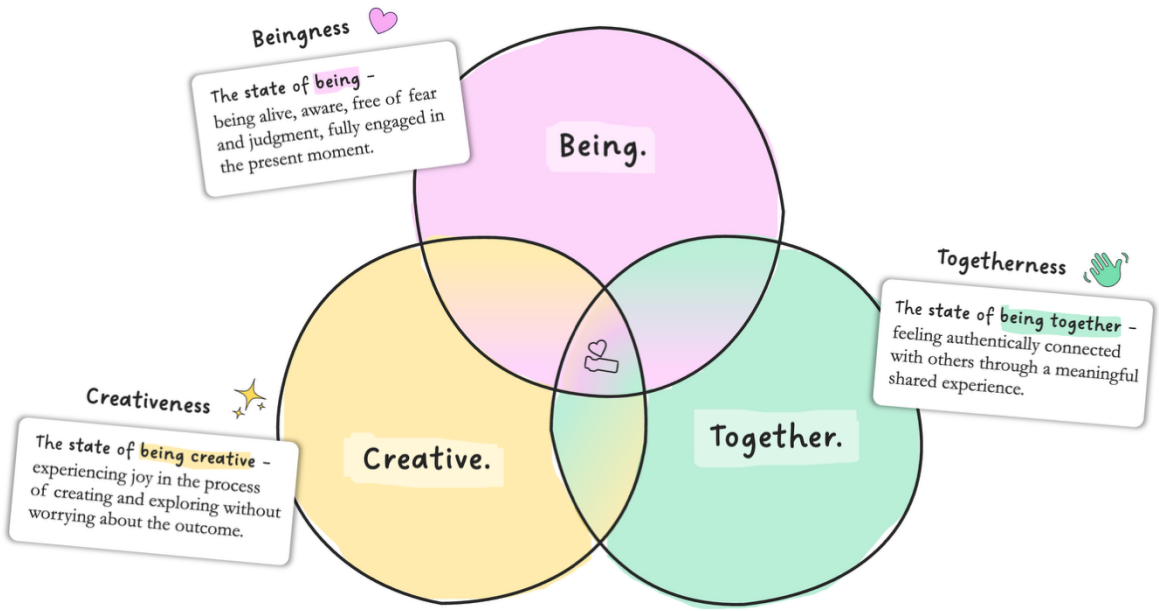


Fig 4: Being. Creative. Together.

In this framing, there is a subtle but significant distinction that we want to highlight. If we were to say that we are designing tools for *creating together*, the emphasis falls on the extrinsic action, outcome, or affordances. In contrast, when we say we are designing for *being creative together,* the focus shifts toward the intrinsic **state** or quality of the experience, one that aligns with the idea of "flow" as described by Csikszentmihalyi:

"The ability to control the environment is one side of the flow experience. The other side, paradoxically, is a feeling which seems to make the sense of control irrelevant… At the height of their involvement with the activity they **lose a sense of themselves as separate entities**, and feel harmony and even a merging of identity with the environment." [18]

While the concept of "flow" often refers to an individual creative experience, we wanted to extend the idea to explore how we can design environments that help facilitate "co-flow" experiences for young people—where they can feel free to express their full self while simultaneously losing a sense of separate self, feeling harmony *with others*. This is the state of *being. creative. together.*

Part 2

Co-create

| technologies and possibilities |

"If you want to teach people a new way of thinking, don't bother trying to teach them. Instead, give them a **tool**, the use of which will lead to new ways of thinking."
— *R Buckminster Fuller*

In this section, we highlight our ideas and approach toward embedding the core values of "being. creative. together." into the design of CoCo, a software platform that lies at the heart of this work, and introduce the novel "co-creative" paradigm for real-time digital collaboration that underlies CoCo.

Ch 5.	A glimpse into CoCo: coco.build	28
Ch 6.	Building a new "co-creative" paradigm for digital collaboration	31
	6.1 Designing for togetherness over collaboration	33
	6.2 Knobs over switches: Multiple ways of being creative together	34
Ch 7.	New paradigm, new possibilities: Insights from pilots with children	41
Ch 8.	Extending creative tools into co-creative environments	49
Ch 9.	Designing decentralized communities: Privacy, safety, cultural context	51

Chapter 5

A glimpse into CoCo:
coco.build

Fig 5: CoCo Platform Demo. (2022). YouTube. https://youtu.be/vvWt0rjco0w

CoCo (coco.build) is a new general-purpose "co-creative" learning platform that we have jointly conceived and developed as part of this research [19]. It is designed to empower children and teens to build and collaborate on an endless variety of creative projects *with their peers in real-time*—regardless of whether they are sitting next to them in a classroom or connecting remotely across continents. It supports multiple expressive mediums including creative computing (building on Scratch 3.0 and p5.js [20], digital art, writing, and others. Educators (or parents) anywhere can start their own private communities on CoCo—creating a safe online environment for young people to engage in live co-creative experiences with their friends. We invite you to watch the short YouTube video above to see CoCo in action.

Why the name CoCo?

By the end of Spring 2019, after the experience of the mindfulness retreat and the workshop with children and parents, our conversations had increasingly started featuring words starting with the letters 'co,' such as connection, co-creation, context, compassion, communities. In one of those conversations, the name CoCo stuck with us. It also nicely captured the core idea of 'Connection + Construction.'

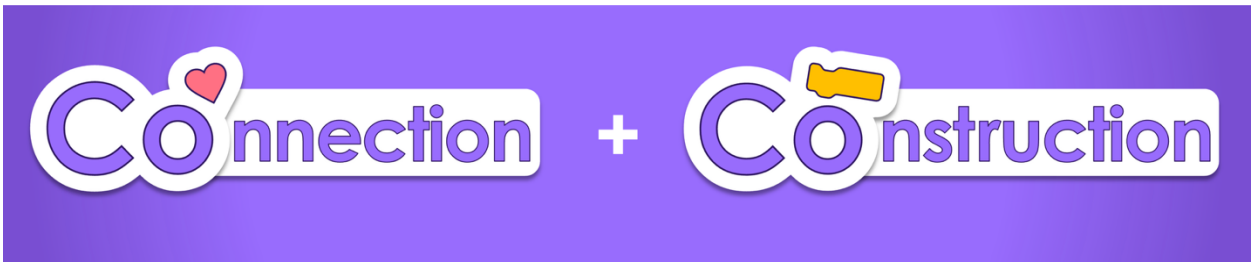


Fig 6: Connection + Construction

> Fun fact: The *.build* domain (of our platform coco.build) is meant for businesses and individuals working in the *construction* industry. We thought it was fitting for CoCo and registered it right away. Looking back, we are glad we did.

Over the course of next few months, while we were engaged in other projects and initiatives, we continued to be drawn to these ideas, and kept refining our thinking and sketches about how we can design a computational environment for young people anchored in the core values of *being. creative. together*. We had an initial working prototype ready at the start of 2020 and got very positive feedback during the first playtest with the members of our research group.

But, creating something *with love* can take time. Imagining and building CoCo has been a long and continuing journey of heart and hard work—packed with countless iterations, choices, an endless variety of software bugs, sometimes too many ideas and too few hands, and sometimes... a pandemic that put everything on pause.

We introduced CoCo to the world in a tiny way in December 2022—sharing a short video and a link to the website in a tweet—inviting people to sign up for beta. We didn't know what to expect. In the coming days (and months) as more educators and

communities came to know about it and shared it ahead, it was deeply satisfying to see the resonance in their reactions.

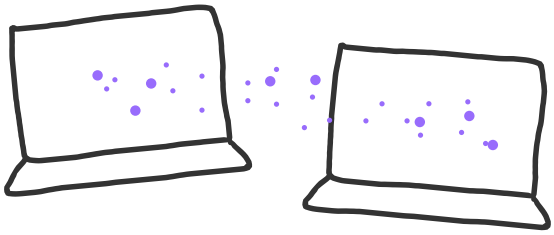
- This looks absolutely brilliant… opens up so many possibilities."
- "Ever feel like you got a glimpse of something that was about to change things?"
- "We hadn't been this excited about a youth platform in a long time."

Fig 7: Response from educators and communities

In the next chapter, we dive into the design of a new "co-creative" paradigm for digital collaboration that we introduce through CoCo, and highlight how it opens up a new space of possibilities which have so far been unexplored.

Chapter 6

Building a new "co-creative" paradigm for digital collaboration



Many of the existing creative online environments for young people, such as Scratch and others, are not currently designed to support real-time collaboration with peers. While the dominant assumption might be that collaboration is primarily useful when people are physically apart, like in a remote learning context, we recognized that regardless of whether learners are digitally together on a video calling platform like Zoom, or physically together in a classroom, they remain creatively distanced from one another.

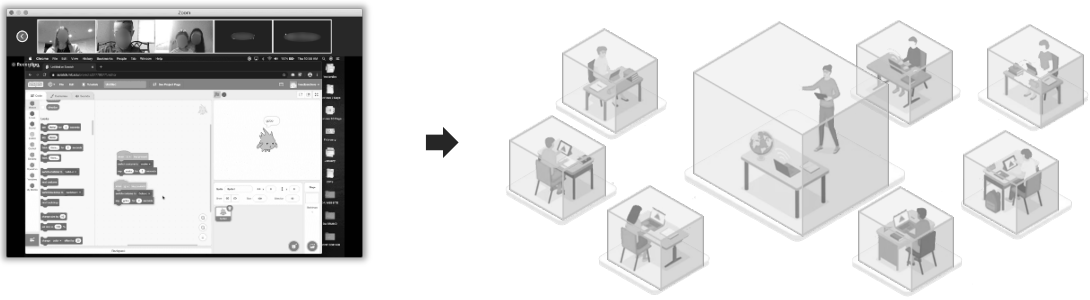


Fig 8: Digitally together but creatively distanced

For instance, on a video call, the experience often evokes the visual of learners (or educators) being stuck in glass boxes. They can see and hear each other, but cannot engage in any meaningful creative interaction with one another.

When it comes to physical spaces, even when a space may be designed to foster more interaction between learners, such as by having learners sit in pods instead of rows, the creative learning tools that young people use typically support only individual work. Children therefore largely work on their own projects on their devices and are not able to co-create with their peers in a shared digital space. The educator also has no visibility into their creative process and often has to stand behind a student to peek into their screen—which can cause some learners to feel awkward and conscious.

Fig 9: Image from booleangirl.org/clubhouse

Existing paradigms for digital collaboration

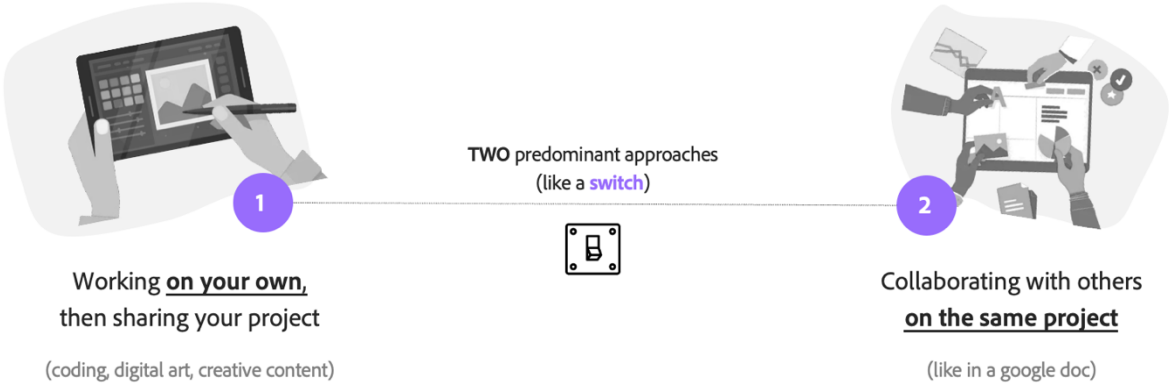


Fig 10: Two modes of online collaboration

Existing digital creative tools typically employ one of two predominant design paradigms toward integrating social interaction or collaboration with others:

- Tools for working individually on creative projects and then sharing the completed works with others in an online community (such as Scratch).

- Tools that support the Google Docs approach of collaborating with others on the same project (such as drawing on a shared digital canvas, making edits to the code in the same project, or writing together in the same document).

Collaboration doesn't always work well for everyone

There is no denying that collaborating with others can be a really joyful, productive, and energy-giving experience, but it does not *always* work well for *everyone* involved. The reasons can be many—not being able to learn and work at your own pace, not feeling comfortable to express yourself in the group, feeling or being judged, someone hogging the work, perceptions of unfairness in contributions and credit, and other negative feelings that can sometimes arise out of this experience.

> Computer interfaces should give us new abilities, useful and interesting enough that they would be used not only when people were physically apart, but even when they were able to be together. — *Hollan and Stornetta [21]*

6.1 Designing for *togetherness* over collaboration

In designing CoCo, we took a different approach—consciously centering togetherness over collaboration. We chose to design the digital environment in CoCo as a shared space for co-creating projects in real-time and not necessarily for collaborating on the same project in real-time. Collaboration often exclusively implies *working together*, while a shared space more broadly implies *being together*. It is hence inclusive of collaboration as one of the ways of engaging with others but not the only way.

In envisioning the interface and affordances, we felt inspired to capture the setting and spirit of a creative art/craft workshop space in action: a group of children sitting together around a table, making their art projects. They can peek at each other's creations and get inspiration; they can easily exchange craft materials

Fig 11: Group of kids creating art projects in a space

with one another; and, if they want, they can choose to collaborate with anyone in the space. This is a similar spirit evident in the Reggio Emilia schools in Italy, or in Computer Clubhouses [22].

6.2 Knobs over Switches: Supporting multiple ways of being creative together

Through CoCo, we introduce the design principle of *knobs over switches* to reimagine the space in between the two extremes or switches (of working on your own and working together on the same project) as a new kind of shared digital space that captures the experience of togetherness. Just like how a knob affords finer control over one's preferences, similarly, a CoCo space is designed to provide creators with the agency to choose when and in what way they'd like to engage with their peers, based on their own comfort and preferences at any moment.

Here is a spectrum of the different modes young creators can use to engage with peers in a CoCo space: 1) Shared Presence; 2) Playful Inspiration; 3) Fluid Recomposition; 4) Creative Interaction; and 5) Coordinated Collaboration

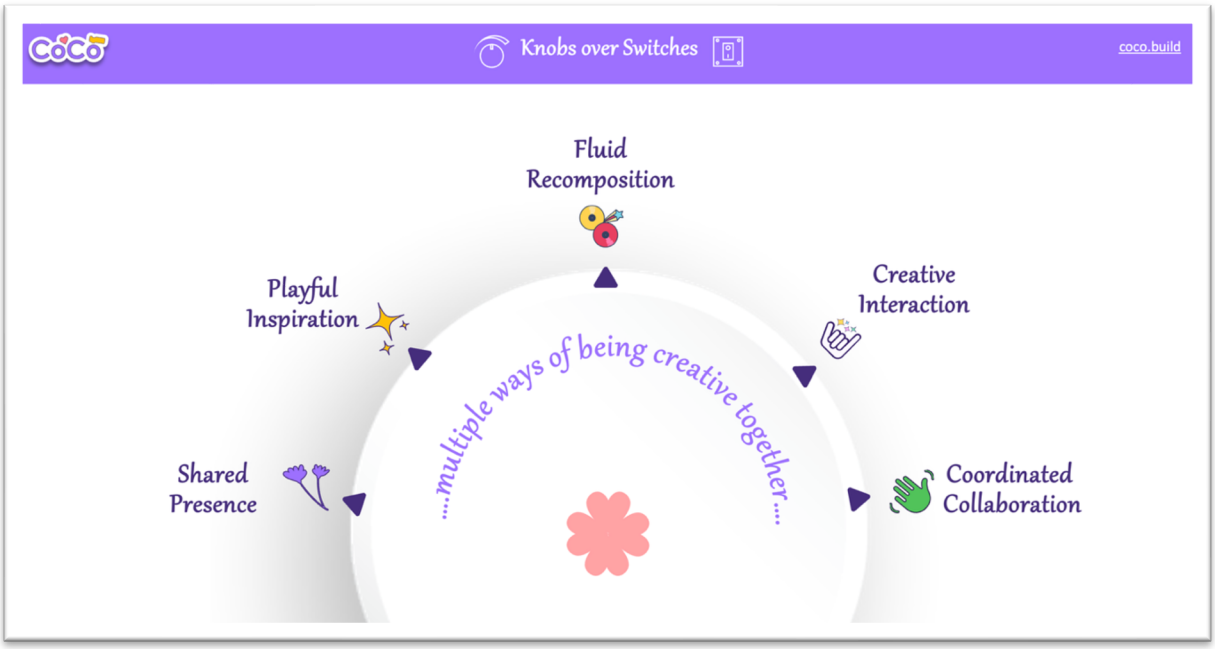


Fig 12: Knobs over Switches: Multiple ways of being creative together

Next, we describe each of these modes and highlight the space of possibilities that these open up for young people.

1) Shared presence

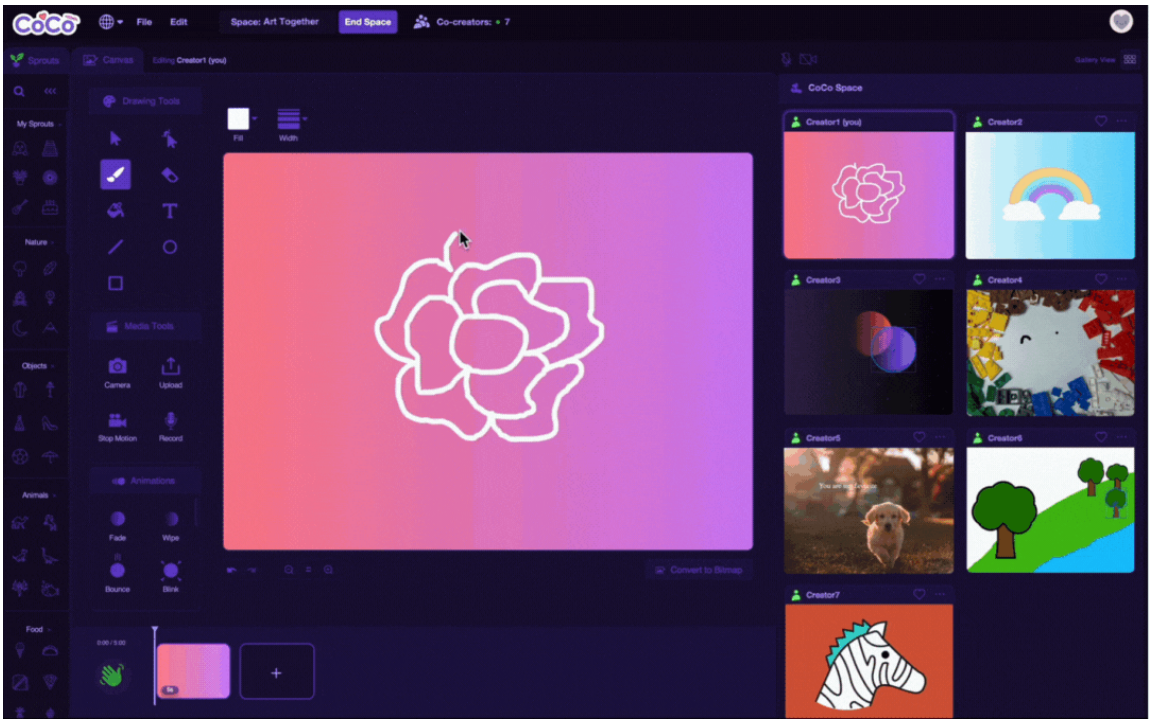


Fig 13: Shared Presence

Context is communication. Just being together and sharing a context with someone is itself a form of communication. In CoCo spaces, creators can work on their projects in a shared digital context, in the *presence* of others. Much like sitting in a café. This affordance is of course very useful for engaging learners remotely, but equally powerful when they are together in the same physical space. Even if students are sitting at a distance within the classroom, digitally, in a CoCo space, they are always right next to one another.

2) Playful inspiration

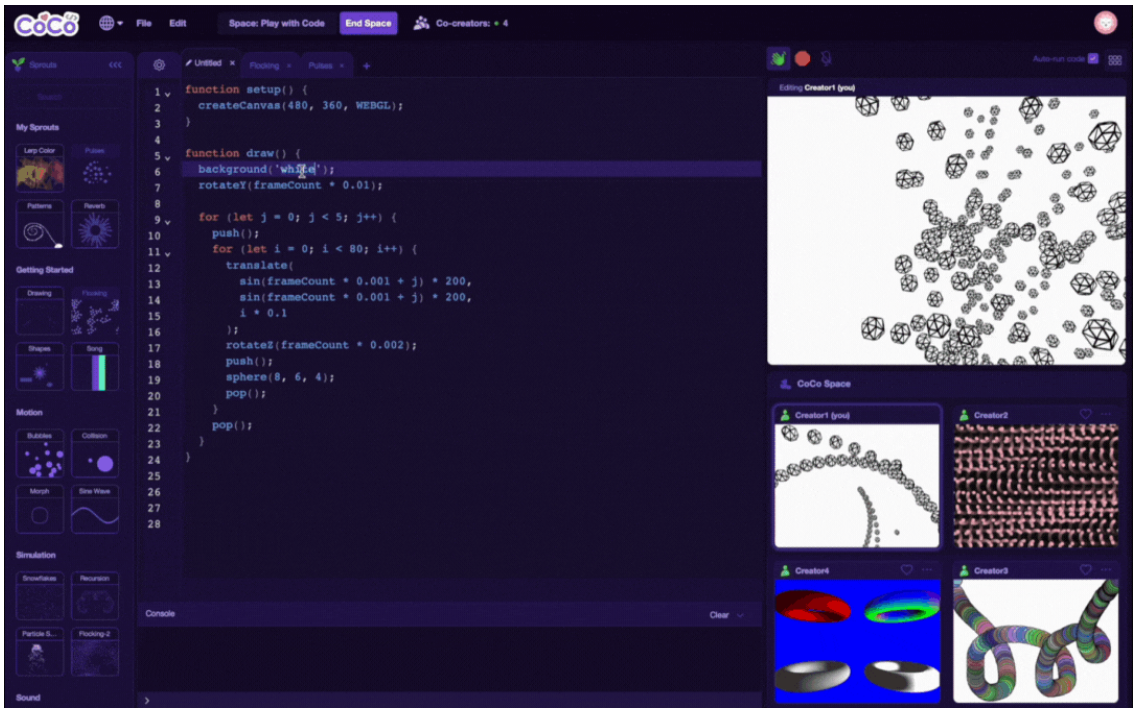


Fig 14: Playful Inspiration

Curiosity often leads to finding inspiration. It can be challenging for young people to get started on a blank canvas in any open-ended creative environment. In a CoCo space, even though learners get to work on their own projects, they are always surrounded by multiple points of inspirations in the form of the mini live windows showing their peers' creative explorations in real-time. They can also freely follow their curiosity and peek into anyone's project to view and interact with it, without having to change their context. Inspiration and support naturally becomes abundant in a shared space like this, making it easy for young people to transform their interests and ideas into creative projects.

3) Fluid Recomposition

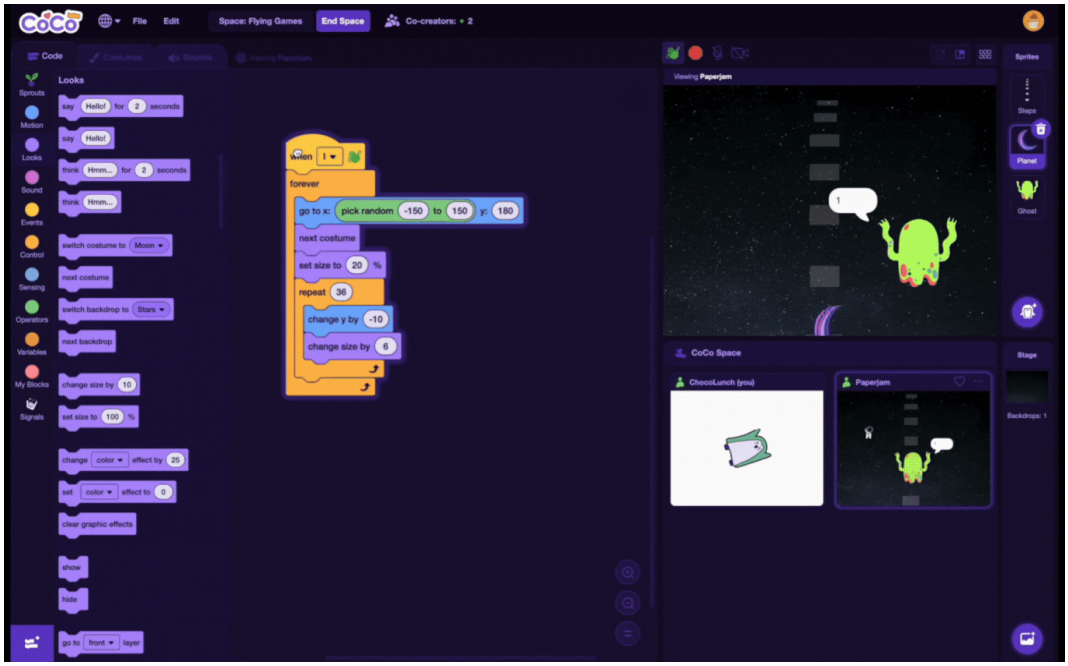


Fig 15: Fluid Recomposition

Creation is fusion. In addition to supporting the fluid exchange of ideas and inspirations *in* the process of creating, CoCo spaces also support live remixing of code and other digital assets between peers. Any creator can drag and drop a piece of code from another creator's project on to their mini window and it will instantly appear in their own project. They can then change it however they like. It is similar in essence to a sight of children sitting around the same table and freely passing and sharing craft materials with one another. Through this active fusion, remixing, and recomposing of one another's creative ideas and work in real-time, young people's digital creations start getting enmeshed with one another in a variety of ways, making a rich space for conversations about—*whose creation is it anyway?*

4) Creative interaction

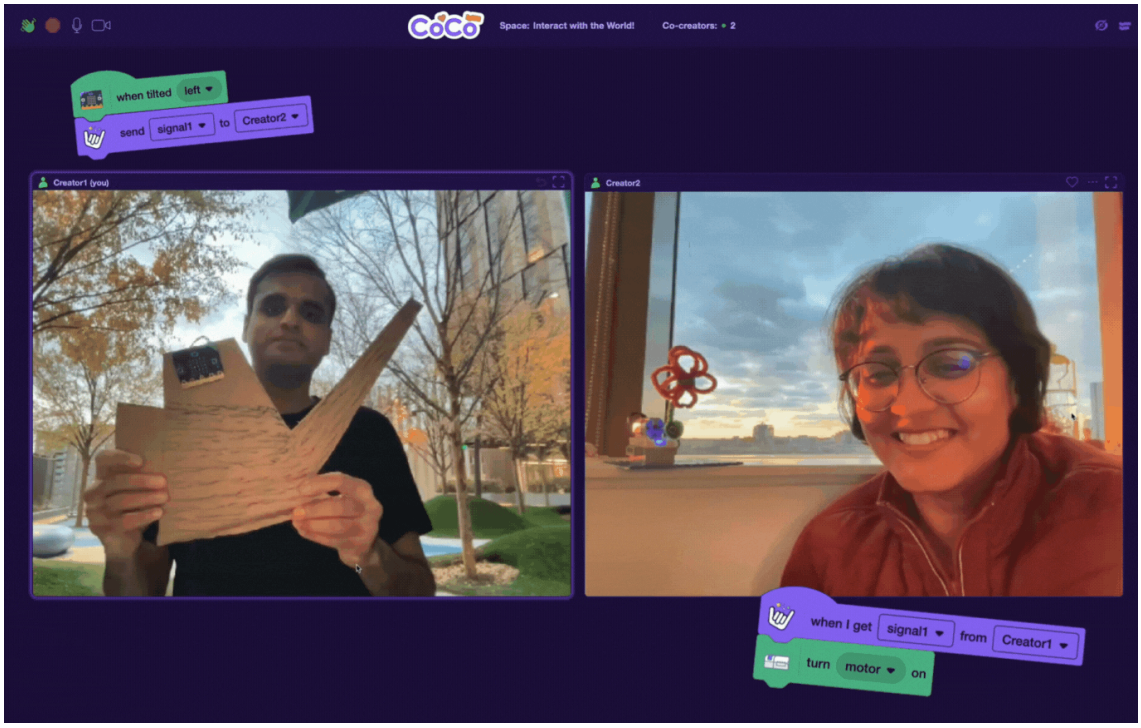


Fig 16: Creative Interaction

Interaction forms a connection. The CoCo Blocks environment introduces a variety of new peer programming blocks—Signals, Waves, and Shared Variables. Creators can use these blocks to make their project trigger something in their peers' projects in real-time and also get to see the outcome instantly. This opens up a world of new co-creative possibilities. They can program multiplayer games, chain reactions, dynamic stories and animations, interactive video-based experiences, and an endless variety of other shared creative experiences together.

5) Coordinated collaboration

Fig 17: Coordinated Collaboration

Co-creation often leads to serendipitous collaboration. Having real-time access and visibility of your peers' works in progress is conducive for sparking organic forms of creative collaboration between groups of young people. In addition to supporting *collaborative projects* where creators can contribute different parts to a single project, CoCo spaces also afford building new types of *collaborative experiences* as a group. For instance, creators can collectively imagine and program a new kind of multiplayer Pac-Man game where the character passes through each of their projects turn by turn and they can use shared variables to have a common score! Or, they can code their own musical instruments and then play their projects together in the form of a musical band. There are countless possibilities.

The underlying goal behind all of these design choices and affordances is to cultivate a joyful co-creative environment that is inclusive of young people's diverse interests and personalities and their multiple ways of being, of learning, and of engaging with others.

Reimagining 'live' online learning with CoCo: Empowering educators to support learners better, wherever they are

The term, *live learning* almost invariably evokes an image of an educator and a group of young people engaging on a video calling platform like Zoom. The educator will be seen doing a screen share of a platform that they are using (learning tool, coding tool, design tool, etc.) and often, not surprisingly, the kids will seem awkward and uncomfortable sitting in front of the camera. This *passive* mode of engagement has ironically come to be called as *live learning*—when in reality, most kids are neither able to feel a*live* nor *learn* in an environment that literally translates the experience of a traditional classroom to a digital context.

We think the built-in video and audio functionality in CoCo—although very useful for supporting remote experiences—is not what truly makes the environment *'live'*, rather it is the ability to engage in a wide variety of creative interactions with peers in real-time *in* the process of learning and building. CoCo spaces are therefore designed to empower educators to support children better *regardless* of whether they are physically together in the same room or connecting remotely from their homes. Moreover, video in CoCo is not a *passive* medium, it is a programmable material that young people can tinker with and use to build new kinds of *shared* creative experiences with their peers.

Chapter 7

New paradigms, new possibilities:
Insights from pilots with children

I think that CoCo is... AMAZING!!!!!!!!!!!!
😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍
- Isae, 10-yr-old

I've never seen a real-time collaborative space like that!
- Educator, Germany

Oh, here you are **making and sharing** a project at the same time - **not as separate steps**!
- Luca, 9-yr-old (who loves Scratch)

Co-creating, as we've seen over the years, is essentially a lot of co-iterating. The process of designing and iterating on CoCo has been research in itself.

Over the past two years, we've engaged **over 2,000 young people** in exploring CoCo through our pilots and playtests across diverse contexts. These involved both in-person workshops that we conducted as well as sessions facilitated independently by educators in formal and informal settings across multiple regions, including in *United States, Canada, Brazil, Colombia, Dominican Republic, Netherlands, Spain, Turkey, and others*. We have continued to learn and iteratively refine the platform based on these experiences and feedback.

Young people (ages 8-16 years) have created more than 1600 CoCo spaces on the platform, with each space having 2 or more co-creators. While it has been exciting to see all the wonderful new co-creative possibilities that young people have explored using the platform—we think the **most meaningful insights come from young people's own reflections about their experiences**.

This chapter is therefore not an evaluation section in the conventional sense. For us, this part of research isn't meant to be a way to "prove" the effectiveness of this new paradigm, or even a way to just improve the features of the platform. It is a medium for engaging young people to reflect on what it means for them to engage in co-creative experiences. In framing the open-ended question(s) for the post-workshop feedback form—we always thought about whether it's a question that will be meaningful for children to reflect on.

We deeply resonate with Reggio Emilia's approach to capturing and sharing insights about children's learning which stems from the fundamental question they keep at the center—"How can we help children find the meaning of what they do, what they encounter, and what they experience?" [23]

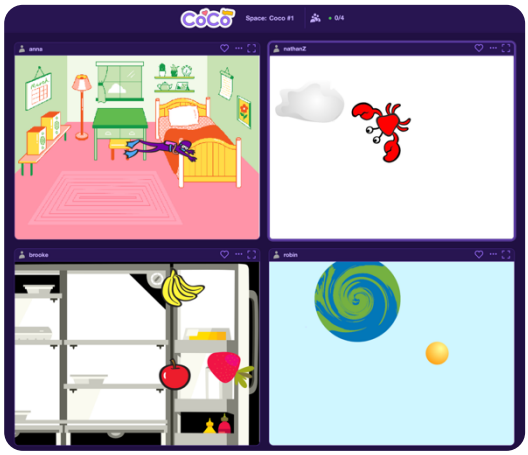
In that spirit, we interweave children's explorations and reflections to highlight some of the themes that stood out to us and that we hope can help spark others to design tools and environments that can support children in co-creative experiences.

- Agency in being creative together

Throughout our workshops, we observed young people naturally moving along the spectrum of engagement options provided by the "knobs over switches" design paradigm in CoCo (shared in the previous chapter). Some jumped right into collaboration, while others started off independently and gradually moved toward collaboration, and some others who enjoyed just working on their own alongside others in the same space. What stood out was how children appreciated the variety of ways they could engage in being creative together, while still having agency and choice in how they participated.

"

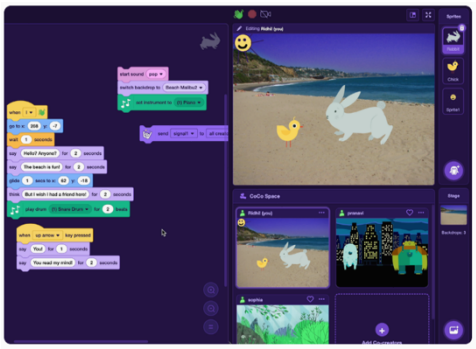
It made it versatile. **I could work as independently or collaboratively as I wanted**, and at any point I could still check in with my peers. It was interesting to see how my peers and I developed different ideas. **It was cool to see the creativity flowing**. And made the process creative, fun, fulfilling, and intellectual.
— *Anna, 15-years-old*



- The invisible co-imagining in action

Often, if we only look at the final projects or even the data, we might not notice any direct remixing of code or obvious similarities between the projects in a space. In the example below (and in many other examples), we saw this kind of invisible co-imagining taking place, which often isn't evident in the output itself, but becomes clear through children's reflections. It suggests how inspiration naturally flows when young people have visibility into each other's creative processes in real-time.

“



One person added sound in their code when I forgot to add sound in mine, so I did that, and that same person made a conversation between two sprites, so I did that as well. **It was nice to see other people’s projects as they sparked my interest to add things to mine**.
- Ridhi, 12-years-old

- Serendipitous collaboration

One powerful moment came during a workshop we had organized at MIT, where young people from across the country had come to participate. Normally, as we were designing the platform, we had thought that the mode of building a unified collaborative creation in CoCo would likely be explored more in spaces where young people know each other already or are friends with each other—like in a classroom. But in this setting, four girls (ages 12 to 15) who didn’t know each other but were in the same digital space in CoCo, spent over two hours together imagining and creating a fun detailed collaborative story. We asked one of them about the experience:

“

When we started making this project, I feel like we didn't really plan on it like becoming this sort of movie. We just like started creating our own background and then what sprites we wanted and then after we saw like what each of us had made, we thought we could just connect all the scenes. So like, if you saw the different sprites traveling to a different scene and then we could, like, connect it like that. And add

some silly dialogue and action and then kind of like make it into a movie.
—*Hannah, 15-years-old*

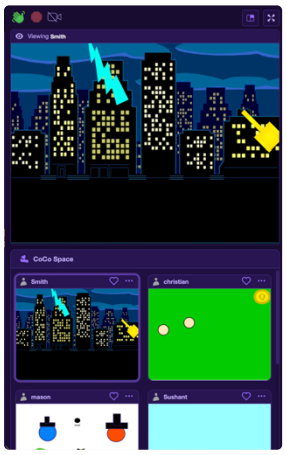
It made us see reflect on how **the design of the tool can enable creative collaboration that otherwise might have been unlikely.** In this case while they started initially working on their own projects in the space, they eventually decided to combine them into a story. After the workshop, they shared how they have all become friends now. It's nice to see how technologies can be utilized to serve as a medium for creating meaningful connections.

- Children as facilitators

of each other's learning and creative processes in real-time...

Often, in classroom settings, there's one educator facilitating a big group of young people. Everyone is largely working on their own device. While some young people are shy to ask for help, others might have a very specific doubt that the educator may or may not know in the moment how to help with. It can sometimes take away time and attention from others in the room. It is unfortunate that in a space where a group of young people are sitting together, the tools often do not allow them to leverage their collective creativity.

Through our workshops, we observed how a co-creative digital environment can empower children to naturally became facilitators of one another's learning and creative processes in real-time. We consistently heard from children how both asking for help, offering help, and accepting help in the process became so easier for them:



"

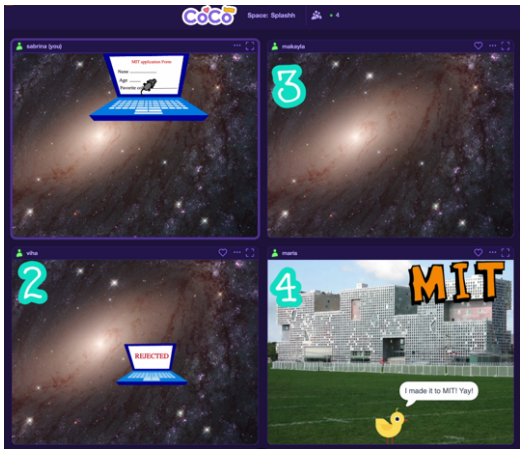
I think since I was able, since everyone was able to see each other's code, I definitely **found it easier** so that when they had a problem, we could identify it quicker and all of us could like, **look at each other's code** and then we would go over and like **help the other person with that issue**.

“

I remember when I was trying to figure out how to get a certain movement option for one of my sprites and **I looked at my partners project and was able to copy it and understand it from their project**.

- Supporting young people's diverse personalities and preferences...

Collaborating with others often is particularly hard for people who are not naturally social or extroverted. Often when an educator makes groups in classrooms, asking students to collaborate together, some young people may particularly find it hard to find this a positive or joyful experience. While using Scratch in a classroom, for instance, a teacher might ask students to work in pairs on a single project by sharing a device. One person might take more control and the other person might not feel comfortable or free to contribute, or both might wrestle over to take control of the device. In CoCo, young people can work on their own devices while being connected in real-time in a shared digital space with their peers. They can see one another's projects in the same digital context and choose to collaborate in their own ways. The collaboration is not forced upon but emerges naturally.



In this memorable project from one of the workshops, a group of kids—many of whom had just met—created a whimsical collaborative story about a chick from the galaxy who applies to MIT, gets rejected, and then somehow crash-lands into MIT anyway. They recorded their own sound effects and voices, layering humor and imagination into their project. When asked about their experience, one of the participants reflected on how they thought the platform supported people with diverse personalities:

“

..this is kind of a way that **whether you’re introverted or extroverted, you can still work with others** and like you all get an equal amount of work to do on the project, if that makes sense. Like you’ll all have an equal share of what to do, and **it’s not like one person’s left out and another person takes on all of the work to do**. And you can create separate projects. You can combine them all together.

- The affective dimension of co-creation

The “co-” in co-creation represents connection with others *in* the process of creating. Across all our workshops, young people have shared how being able to create and work together with their friends is something they really enjoyed in CoCo.

Unsurprisingly, many children’s comments about their experience on CoCo reflected this emotional and affective dimension of co-creation. Often kids would say, “I loved collaborating with a friend,” and describe how it was really fun for them to get to co-create together in real-time. But it was also interesting how this sense of connection emerged even among kids who didn’t know each other beforehand. These reflections highlight how meaningful authentic connection with others can be a powerful motivator in creative learning—supporting not just what children make, but how they feel while making it.

“Creating alongside others in the same space made the experience **more fun and motivating**.”
“It felt **more nice**.”
“I felt **more motivated**.”
“It made it **less isolating**, more of a collaborative environment.”
“It affected my experience positively; I was able to collaborate effectively with my peers and make new friends. **I felt happier** than when I worked on my own, like in Scratch (not that Scratch is bad, I love it as well!).”

There are so many more examples, comments, insights, and experiences we could highlight. But overall, through our playtests and pilots, we continue to see how children naturally seek and enjoy many forms of creative connection—from just creating alongside others to directly collaborating—and that digital creative environments can and should support this full spectrum rather than limited choices.

Beyond the features and outcomes, ultimately, we should design for co-creativity because creating with their friends and peers makes children feel *happier* than creating on their own.

Chapter 8

Extending creative tools into co-creative environments

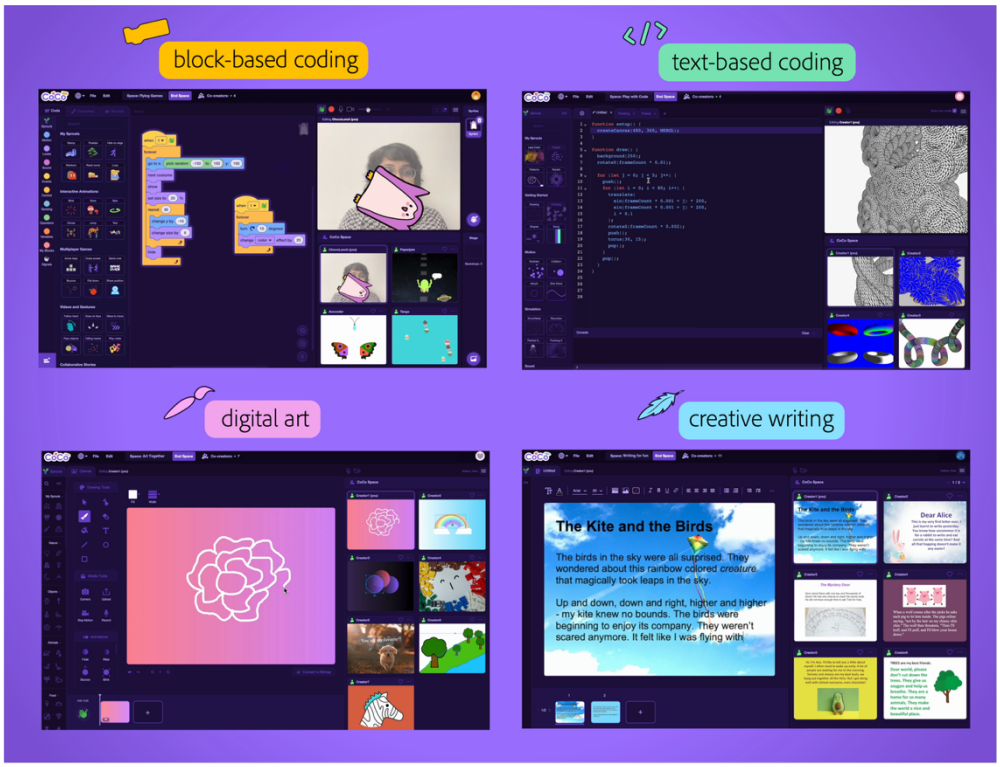


Fig 18: Multiple co-creative environments in CoCo

We've especially designed CoCo as a general-purpose co-creative platform—one that supports collaboration across a wide range of creative environments such as Scratch, p5.js, and others. From the start, we've always envisioned CoCo not just as a platform, but also as a paradigm. Our interests and explorations have not been toward building a better coding platform, but rather to foster a new paradigm: where young people can choose from multiple expressive mediums of their choice and engage in real-time, co-creative experiences with peers in a shared digital spaces.

As AI continues to expand what's possible—from intelligent assistance to code generation or creating images—we're excited to explore and integrate these new affordances. But regardless of advances in these directions, the need for authentic co-

creative engagement remains essential. We continue to keep that at the center of all our explorations. The social, emotional, and relational dimensions of being creative together will only become more important in this new landscape.

We think this paradigm can extend to an endless variety of digital creative experiences and environments. For instance, we're already prototyping CoCo Canvas for co-art and exploring co-writing environments. In CoCo Canvas, young artists can make art on their own canvas in a shared space, they can easily grab and remix parts of one another's artworks in real-time, get inspired by others' color choices or styles of drawing, or collaborate to make a group art piece.

These affordances of engaging with one another in real-time will continue to be exciting and valuable even when all creators have access to AI assistance as well. There is an endless variety of amazing creative environments that talented groups of people everywhere continue to build, and we imagine and hope that at some point in future, CoCo can host an open library of a growing number of community-sourced co-creative environments for educators and young people to choose from every time they create a CoCo space.

Chapter 9

Designing decentralized communities:
Privacy, safety, cultural context

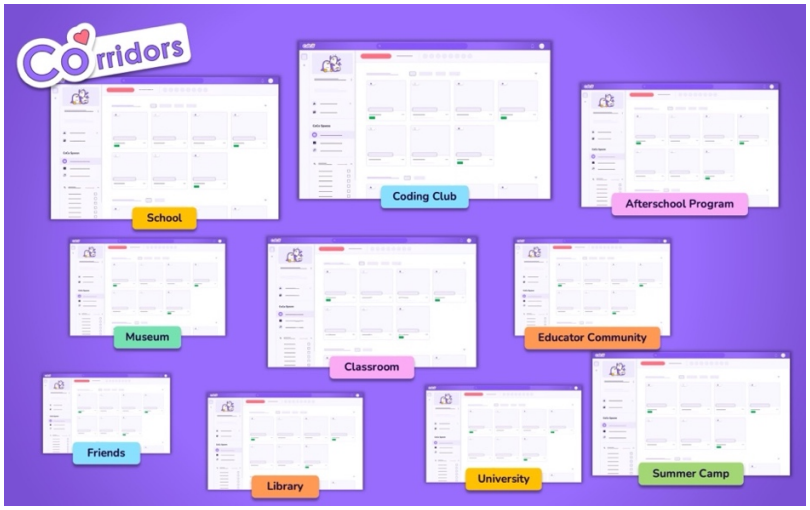


Fig 19: Decentralized private communities in CoCo

Since CoCo offers real-time co-creative environments for children, we have carefully designed the surrounding social infrastructure of the platform in ways that prioritize children's safety. Instead of a large open community (or open online game environments), CoCo is designed to let educators create and moderate their own private communities. We call them "corridors," inspired by MIT's famous *Infinite Corridor*—an 825-foot-long hallway that runs the entire length of MIT's main buildings and connects to the different classrooms, lab spaces, and social spaces for members of the MIT community.

Fig 20: MIT's Infinite Corridor

Similarly, educators can invite students from their classroom or after-school club to their corridor, using a secure login flow that does not collect any personally identifying information. The corridor provides a safe context for young learners to create and engage in any number of live CoCo spaces with their friends and classmates. They can also explore all of the past spaces and interact with the projects that were created by their peers and even remix a whole space if they like.

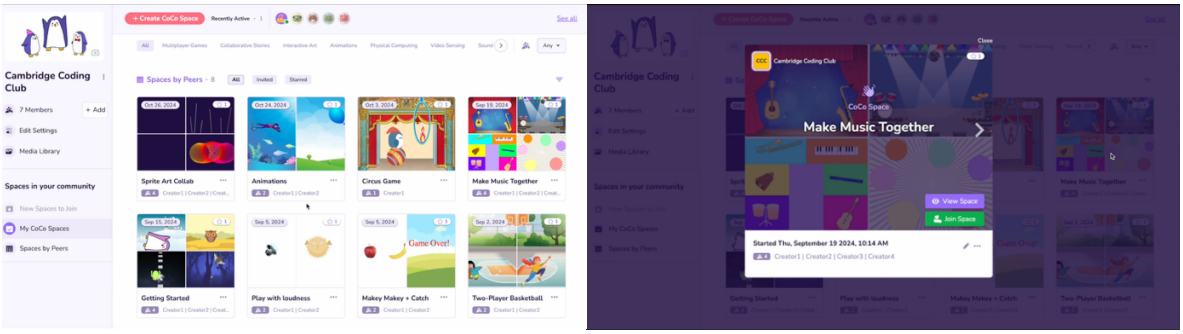


Fig 21: Decentralized private communities in CoCo

Educators can also add custom assets (such as images, sounds, sprites, etc.) in their corridor's media library. Any assets they add are automatically made available inside the CoCo environments. Children can then use those to co-create projects that closely reflect their surroundings and cultural contexts. Educators and researchers are increasingly highlighting the need for supporting young people in collaborative and culturally relevant computing experiences [24]. We hope these design choices and affordances can help empower educators everywhere to support learners better across diverse geographical and cultural contexts. We have seen several educators utilize these affordances in our pilots. For example, (Fig 19) an educator in Colombia added images of the school mascot in their corridor on CoCo for kids to use in their projects.

Fig 22: Adding custom assets in CoCo

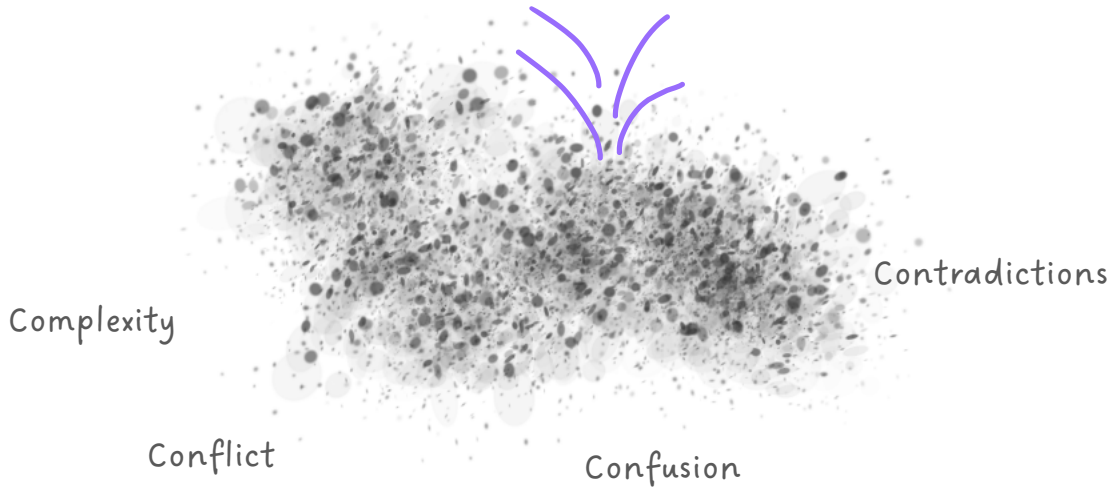
Overall, we continue to hear from educators how they really appreciate having a safe private community for their students and how students are naturally more engaged because they get to create projects together with their friends.

Part 3

Co-clutter

| a necessary interlude |

It's our differences that make the difference.

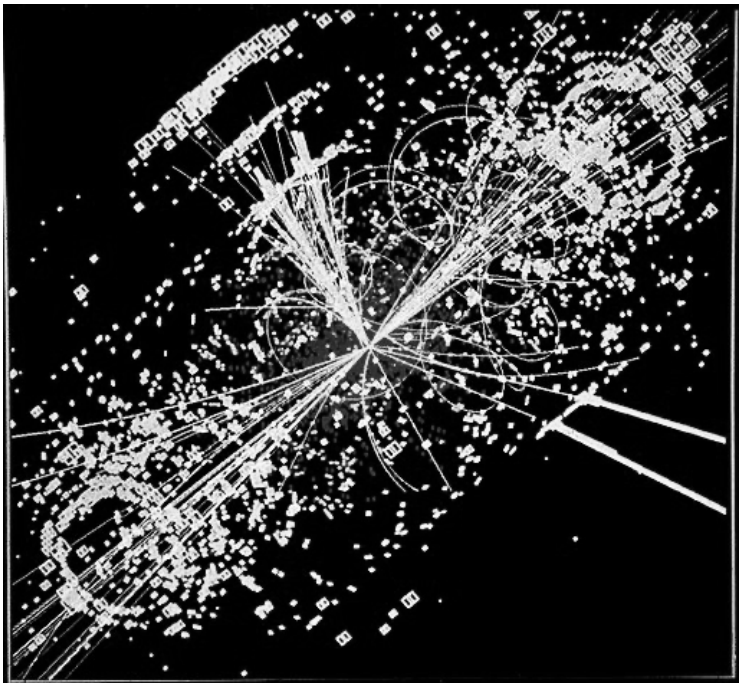


Co-clutter is the flux, co-existing through which, we co-evolve. In this short section, we reflect on how this is an essential part of our co-creative process, and how it is valuable to intentionally design for generative co-clutter.

Ch 10. Creating the conditions for "collision" 54

Chapter 10

Creating the conditions for "collision"



[2]

In the Large Hadron Collider, the smashing together of sub-atomic particles at near light speed produces a shower of new particles.

Interestingly, this imagery closely resembles our own co-creative process. The collision of our different ideas, personalities, communication styles, and moods on a given day can raise the temperature in our conversations. These collisions—sometimes experienced as conflicts—are natural and valuable. The dearer the idea one holds, the more intense the collision.

[2] Image from https://commons.wikimedia.org/wiki/File:CMS_Higgs-event.jpg

Very often during these heated discussions (or after some reflection), we discover new directions, thoughts, and ideas, while simultaneously building a deeper understanding of each other and our own selves.

These conversations are far from negotiations where each person arrives with a fixed agenda. We are never simply trying to "find" common ground—and perhaps this is what leads to exploring and co-creating new playgrounds together. Instead of seeking compromise, we allow the natural friction of our different perspectives to generate unexpected insights.

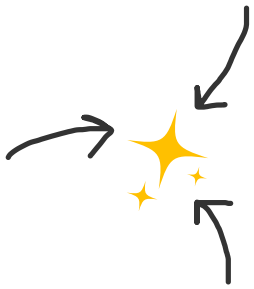
We embrace moments of conflict or co-clutter not as something to be avoided, but as an essential part of co-creation. And over time, we have come to expect, accept, and almost cherish our collisions.

Designing for “collisions”

Seymour Papert highlighted how the role of the teacher is to create the conditions for *invention*.

Through this work, we have explored the value of designing tools, spaces, and experiences that can create the conditions for *collision*. When the diverse ideas, interests, and identities of young people are allowed to playfully collide with one another, new ideas and identities are co-invented in the process.

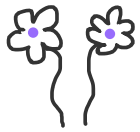
We hope that our work and perspectives can help inspire others to make space for creative collisions in their shared contexts.



Part 4

Co-exist

| shifts and perspectives |



"We have not been able to be secure in human society. There is anxiety of various types in our social existence. You know, we have no anxiety about the way in which the limbs of the body work in relation to one another. You do not go to bed with a feeling of insecurity that the eyes or the ears may not set-up a strike in respect of the stomach which swallows all the food and gives nothing to the eyes or the ears! You have no fear that, perhaps, tomorrow there may be a battle between the right hand and the left hand. But we do not live a life of harmonious organic relationship to our outer life [with others in the society]…Nature is not a machine but an organism. The parts of nature work in spontaneous harmony amongst themselves…"
— *Krishnananda Saraswati*

In this section, we share perspectives on what it means to center the ideas of interconnectedness and harmony in any learning contexts. We illuminate a set of lenses and conscious choices that can help bring a shift in the way we understand and design our creative, educational, and social systems.

Ch 11	From future to present	57
Ch 12.	Constructing *things*, Constructing *context*	58
Ch 13.	Head, hands, heart, and *hüm*	61
Ch 14.	From self-inflating to self-*less* social platforms	63
Ch 15.	Passion, dispassion, and compassion	68

Chapter 11

From future to present

Education is not preparation for life; education is life itself.
— *John Dewey*

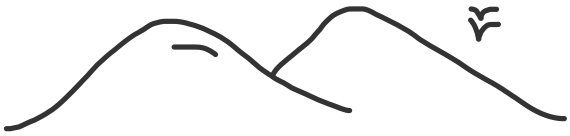
Life can only be found in the present moment.
—*Thích Nhất Hạnh*

Throughout history, the focus of education has predominantly been centered toward preparing children for the *future*. What if we could instead reimagine education as a means to empower children to freely and fully engage with the *present*?

How will we design learning systems, spaces, and experiences for young people if the goal was *not* just to prepare them to be successful in future—where the notions of what 'success' means are narrowly determined by our conditioning? What will we choose to assess, what will we choose to teach, how will we choose to teach—if we shifted our lens to the present?

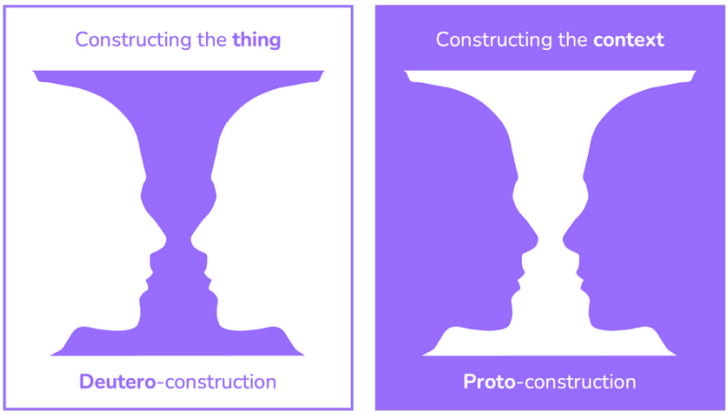
How can we help preserve children's natural ability to stay in the "now" instead of making them fear the future? Isn't that what we constantly strive for as adults?

In the chapters ahead, we explore what it means to design for *presentness*, and the small but intentional design choices we have taken in the tools we are building to help bring this shift.



Chapter 12

Constructing things, Constructing context



> The thing is the thing, the context is the context, the thing is not the context, and the context is not the thing. The thing is being constructed within a context but *maybe constructing the thing is only a context for constructing the context*?

Shifting from Individual and Thing-centric lens to a Co-creative and Context-centric lens

In any shared creative learning space, learners are always engaged in two kinds of construction. The more visible (and seemingly primary) one is the construction of their projects or *things* (such as building a game, writing a story, or creating art) but the other seemingly invisible one is the simultaneous construction of the social environment or their shared context with peers. Contrary to what might seem obvious, we consciously emphasize seeing the invisible constructing of *context* as the *proto-construction* process and the constructing of *things* within this context as a *deutero-construction* process.

We do this reversal with the intention of bringing a shift in the often exclusive and excessive attention put only on the more visible aspects of a learning space (the projects being built, the concepts being learnt…) and not on the shared *context* being constructed. This imbalance typically arises from seeing the learning environments (physical or digital) often just as consisting of *individual* learners / users, each working on their separate *things*.

From two kinds of construction to two forms of learning

> ...every time you learn something you learn two things,
> the other being the model of learning you just used.
> — *Bateson's deutero-learning concept* (as described by Papert [7])

When children engage with(in) any system, they are not just learning the explicit things that the system is designed for (skills or concepts), but they also imbibe the ideas, worldview, and frameworks that the system is built upon.

As designers of these social systems (tools, schools, or spaces), consciously or unconsciously, we end up constructing most of the con*text* or the underlying *text*ure and *text* (rules) of the system *for* children. What does that lead to: instead of empowering children to actively engage in constructing their context, we tend to push children to be constructed and painted *by* the context.

The ideas of success, failure, popularity, etc. are not natural to children but by repeatedly participating in systems that emphasize these ideas, they start internalizing these social structures and eventually start acting from them.

To make this more real, let's imagine there is a hypothetical live online learning environment (let's call it **NoNo**?) that was designed with a typical *individual-centric* and *thing-centric* lens. Here too, children are getting to work on their projects and you can also see your peers' projects in real-time. But also on display are the number of likes, thumbs-ups, and ratings given to your project. In addition, you have the option of sorting all the projects in the space based on these metrics—sort by *most likes*, sort by *highest ratings*, show the most *advanced projects*, and so on.

Also, learners are supposed to work on their own separate projects here, they are not allowed to "copy" or remix anyone else's work.

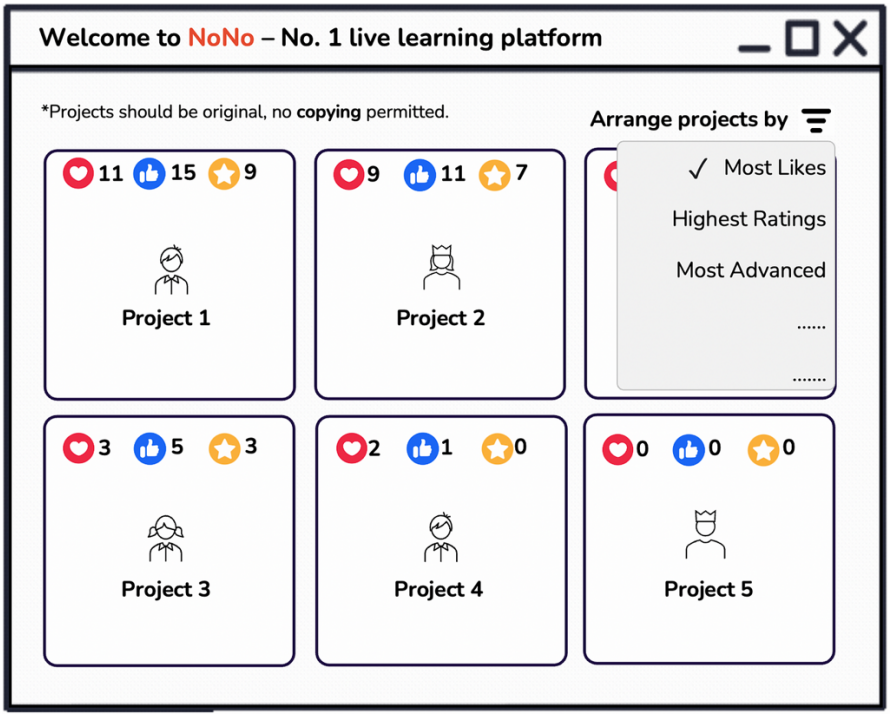


Fig 23: NoNo—A hypothetical live online learning environment designed with individual- and thing-centric lens.

Now imagine yourself as a young person participating in this environment. The environment's context by its very design promotes the ideas of competition, rewards, and comparison. Creators in such spaces are no longer likely to be creating their projects from a place of joy but from an anxious place of excelling in this socially constructed structure.

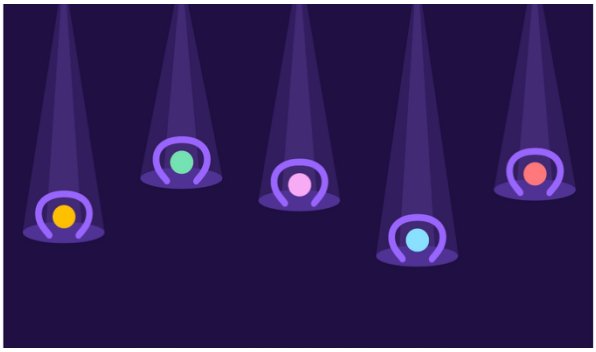
In such an environment, even though children may be *creating together*, they are not *being. creative. together.*

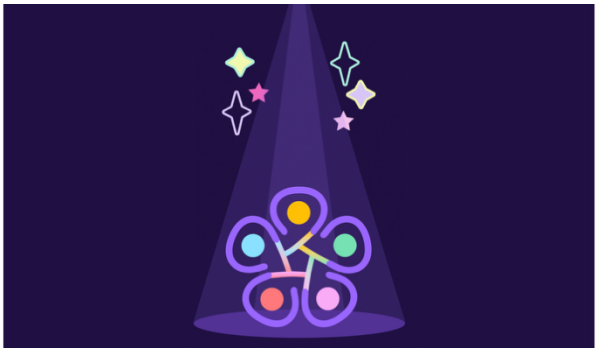
> ..the most notable aspect of a positive judgment is
> not that it is positive but that it is a judgment.
> — *Alfie Kohn [25]*

Can we design spaces rooted in *non-judgment*?

Chapter 13

Head, hands, heart, and hüm





In any learning space, when we start to zoom out beyond the *individual* and the *thing*, and consciously hold the entire space in our view, we'll start noticing the *shared context* that is actively getting constructed, deconstructed, and reconstructed within the space. This shared context is brimming with a rich web of connections that include learners' connections with their peers, with the educator, with their immediate surroundings, and also their connection with themselves.

All of these connections, although invisible, are in fact very real. They are moldable like clay, they are also colorful, they even radiate temperature with varying degrees of warmth to sometimes cold, and they get strengthened not merely by physical proximity but by a deep sense of trust, openness, and empathy in the space. These connections truly come alive in the process of co-creation and if you hear intently, they even make a sound like that of a warm and creatively buzzing space in a state of flow.

We coined a word for it—we refer to it as the "hüm" of the space.

The *hüm* is a representation of the nature and quality of interconnections that are actively being constructed within any social learning environment.

The hüm is what we'll hear, sense, and feel when we take a step further and start to see learners not just as individuals with *head*, *hands*, and *heart*, but also as active co-constructors of their shared context.

The *hüm* doesn't belong to any one in particular yet it is an integral part of everyone, and it contains and sustains the shared values of the space.

If we are not intentional about our choices, we can easily end up designing systems and spaces like the hypothetical NoNo environment above where the *hüm* of the space (which probably will feel more like *h∩m)* will soon start to vibrate with the unpleasant and uncomfortable sounds of anxiety, fear, and insecurity instead of those of joy, connection, and creativity.

Chapter 14

From self-inflating to self-less social platforms

Since the very beginning of developing our ideas, we consciously envisioned and worked toward designing CoCo as a **"self-*less*"** social platform that **departs from the comparison-based, individual-centric paradigm** (of profiles, likes, and followers), which spilled over from early social media platforms to become the dominant approach for online social engagement.

The creator- or self-centric design paradigm is so deeply entrenched in most online social platforms that the idea of a 'self-less' social platform can seem like an oxymoron. It's true that getting opportunities to share content with others online (pictures, posts, creative work, videos etc.) greatly empowers young people's self-expression. However, the inherently performative nature of online social participation, aided by typical social rewards in the form of likes or followers can often deeply influence young people's perception of themselves and their peers in ways that are not always healthy.

The fundamental unit in this model is the 'self' or the user's individual profile where the works created or posted by the user are accumulated, along with the associated social metrics. While it works great for few people who get visibility, for majority of others, this can often lead to feelings of comparison, anxiety, and insecurity, and unnecessary creative pressure. The focus shifts from the joyful experience and process of engaging in a creative process to seeking validation for the final outcome in the form of likes or views.

To live without comparison is to remove a tremendous burden.

—J Krishnamurty

We think any space (physical or digital) is shaped as much by what is present in it as it is by what is *absent*. In CoCo, we have consciously decided to **do away with comparison**. With no individual profiles, no followers, and no likes to collect, there is no digital currency to compare with.

In conceptualizing and developing CoCo, we kept coming back to the question—what underlying ideas, values, and ways of thinking and being will young people discover and absorb as they engage with and within the system? We intentionally made **small and big design choices** keeping that question at the center.

CoCo Cards: Communicating Feelings over Collecting Metrics

Imagine being in a vibrant studio space where you are engaged in making art with your peers. You look over and you really like what your friend has made, you whisper and express your appreciation to them (so as to not affect others in their creative flow). They feel really good and express that to you by saying thank you with a smile. In this spontaneous personal interaction, the likes and the thank you's are not *metrics* to be collected and displayed for others in the room to see. They are just *feelings* that are expressed and felt in the moment. Though this exchange is ephemeral, its affect is often much deeper.

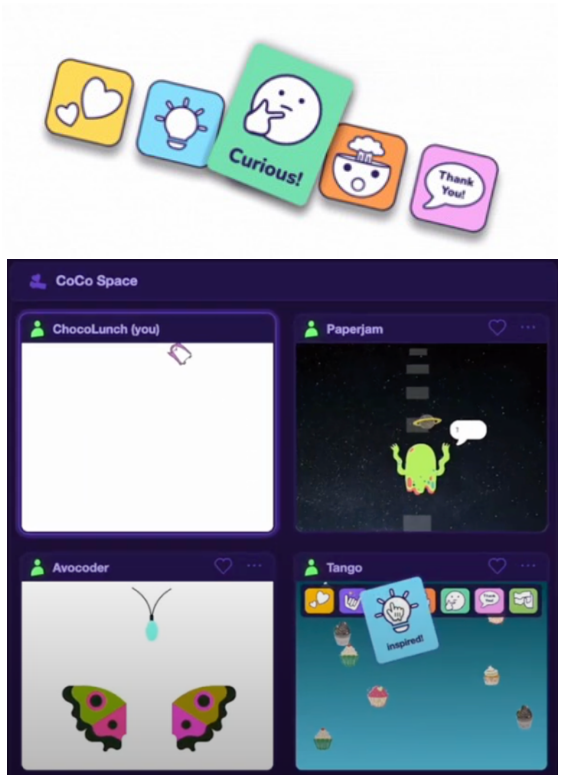


Fig 24: CoCo Cards

We wanted to bring this same personal and spontaneous quality to the social interactions within a live CoCo space.

We designed **CoCo cards** as a way for young people to express feelings of appreciation, wonder, inspiration, or gratitude to one another, as they see one another's projects and ideas evolve in the space. The cards are shared one-to-one

between any two creators within the moment. They are not announced or displayed publicly in the space for others to see.

In our playtests with children, we have often seen children really enjoy and appreciate sharing these little cards with their friends in the space.

Shared Spaces over Individual Profiles

In CoCo, there are no *individual* user profiles or *separate* project pages. The fundamental unit here is not the individual creator, but rather the endless variety of co-creative spaces that they engage in with peers. In CoCo, creators build projects within a shared context that is designed to be inherently interactive and collaborative in nature. Every creator's project is deeply intertwined with other creators' projects in the CoCo space in a variety of tangible and intangible ways. The projects therefore are never separated from the original co-creative context that they were built in. In doing so, we hope to support young learners in making a shift from the often excessive focus that is put on seeking validation for the final *outcome* to instead start seeing the joyful *process* of co-creating as its own reward.

What do young people feel about the choice of NO likes and followers?

In our post-workshop feedback form with over 80 young people (ages 11-16 years), we asked about what they felt about this specific design choice in CoCo is an open-ended question. All of them had used Scratch before and are familiar with Scratch online community built with the paradigm of user profiles, likes, and followers. We knew from our past experiences that this is not a binary choice, and we were curious to hear young people's reflections in their own words.

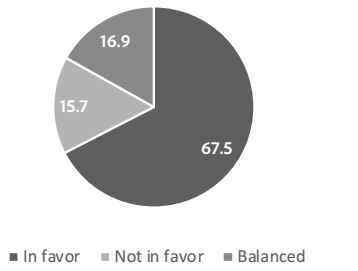


Fig 25: Responses from young people

It was surprising that even when most young people are very used to this

dominant paradigm, in our feedback form, majority of them expressed that they appreciate the choice of removing these metrics. Only about ~15% shared that they prefer these metrics, while another ~15% shared suggestions for taking a balanced approach toward this choice.

Here are some of the comments young people shared in the form:

Expressing - in favor:
"This is honestly a great change because nowadays **I feel like people make projects for farming likes and followers which defeats the whole educational and collaborative purpose**."

"I think this **encourages collaboration over competition**. It's a good thing!"

"I really liked this. I think sometimes that can take away from what creating should be about. Not everything needs to be turned into a social media thing, and instead of feeling pressured to create something others might like, and forgetting that the **point of making projects isn't to get new followers, you can make fun things for yourself** :)"

"I think **it creates an environment where all students are equal** and one doesn't receive more praise than the others and **allows them to focus more on the community part** of their projects."

Expressing – balanced view:
"While I like this choice and think it is very effective in small communities. I think that **likes and followers should be displayed on public projects** so that games with high quality and effort are more likely to get played."

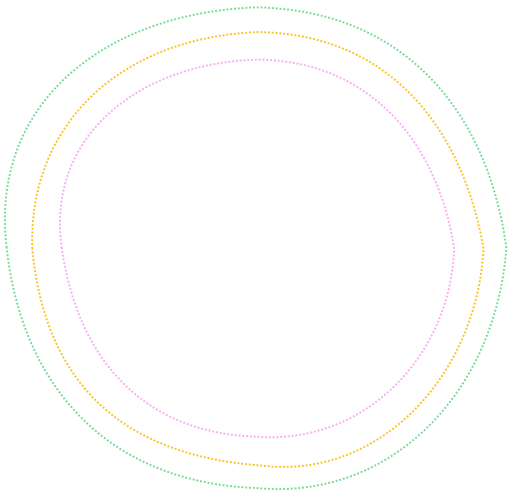
Expressing - not in favor:
"I think that these should be added because **posting your content online for a wider audience can help shed more light on your work**. Likes and follower count just show that people appreciate you."
The point is to not arrive at a right answer. But through this work, we consciously

choose to ask the question, and to help young people reflect on the question—what happens when there's no unit of comparison in a social platform? We think the question is worth asking and designing for.

The word selfless is formed by putting *-less* with *self*, which more literally implies, *free from self*. The part of the *self* we hope children can feel *free from* is the part that gets insecure, fearful, driven by extrinsic rewards, and starts to care only about individual outcomes. **Being selfless is not an act of diminishing one's self, but rather of continually expanding one's self-identity to include the other.**

But we cannot begin to embrace self-lessness as a value when all the social systems we partake in (online or offline) are inherently designed to be self-centric. Instead of bringing us together, they often pull us apart. Ironically, these self-*inflating* metrics—scores, ranks, badges, likes, followers—end up *diminishing* young people's sense of self.

We hope the ideas and design choices we have shared through the work on CoCo can inspire others to explore what it means to design *self-less* social systems and spaces for young people—that serve as playgrounds for young people to continually *expand* their self-identities.

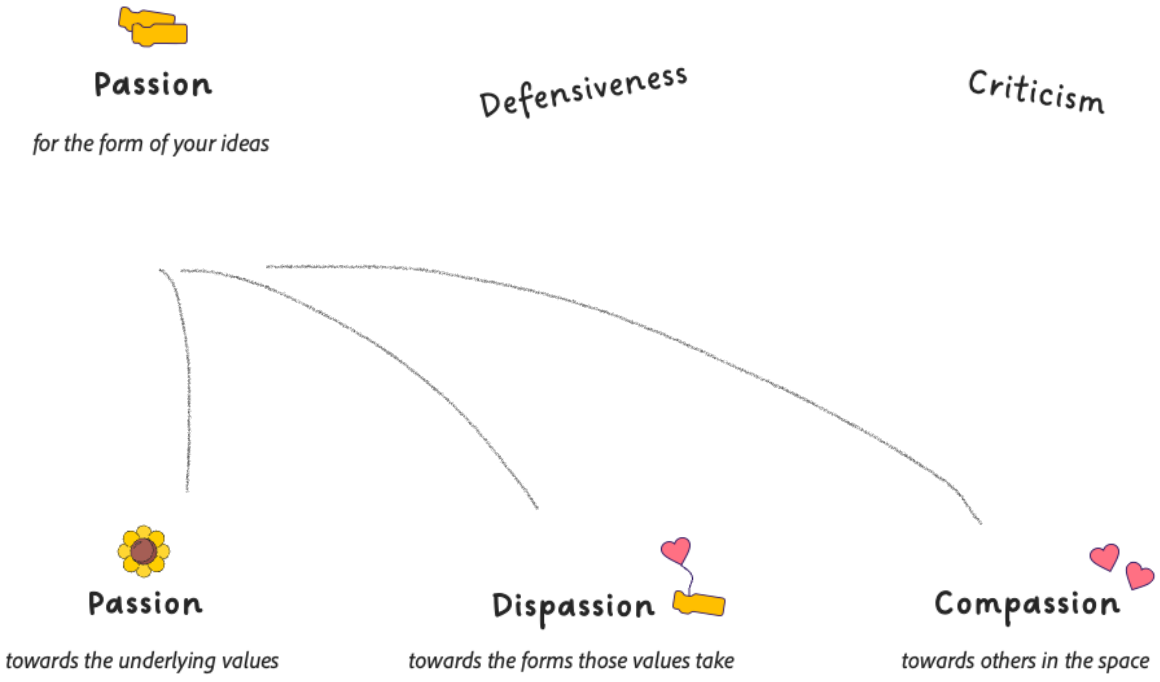


Chapter 15

Passion, dispassion, and compassion

Passion by itself is incomplete, and when it is directed toward the *forms* our ideas take, it can sometimes lead to being overly defensive about our own ideas—and to a mindset of comparison and criticism toward others' ideas.

Passion becomes whole and harmonious when cultivated alongside dispassion and compassion: where passion is directed toward the underlying *values*, dispassion toward the forms those values take, and compassion toward others in the space, and toward your own self.



How can we help nurture young people's
passionate, dispassionate, and compassionate selves?

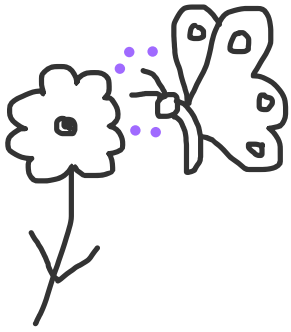
Part 5

Co-evolve

| contributions and directions |

coevolution, the process of **reciprocal** evolutionary change that occurs between pairs of species or among groups of species as they interact with one another.

— Britannica



Ch 16.	The 5Co's of Co-creative Learning	70
Ch 17.	From Monoculture to CoCo-culture	74
Ch 18.	Bringing CoCo to communities around the world	78
Ch 19	Extending CoCo | Little Language Models (LLMs): Empowering children to be future AI modelers	80
Ch 20.	Looking ahead: Centering the *timeless* in timely	92

Chapter 16

The 5Co's of Co-creative Learning

This work is as much about the tools we've built—and the thoughts and ideas we've expressed through words, pictures, songs, and videos—as it is about our shared lived experience and evolving process of *being creative together*. Through living, learning, creating, and of course, colliding together for over 12 years, we've come to recognize five stages that continually shape both our creative work and our relationship. These stages unfold every time we are being creative together. We might jump through these in a single conversation, while also seeing them as phases in our long journey. Interestingly, we've often seen young people in our workshops move through similar stages as they engage in being creative together with their friends and peers.

We now see these stages as part of a process, or a framework, which we call as: The 5Co's of Co-creative Learning—a process in which learning *emerges* through co-imagining, co-creating, co-cluttering, co-existing, and co-evolving—with and **through one another.**

At the center of these 5Co's, are both the *heart and the block*, highlighting how our co-creation of (visible) projects always goes hand-in-hand with the co-creation of the (invisible) context in which they are built. The heart and block are surrounded by a kind of "home"—a metaphor for deep **trust** that holds the process together.

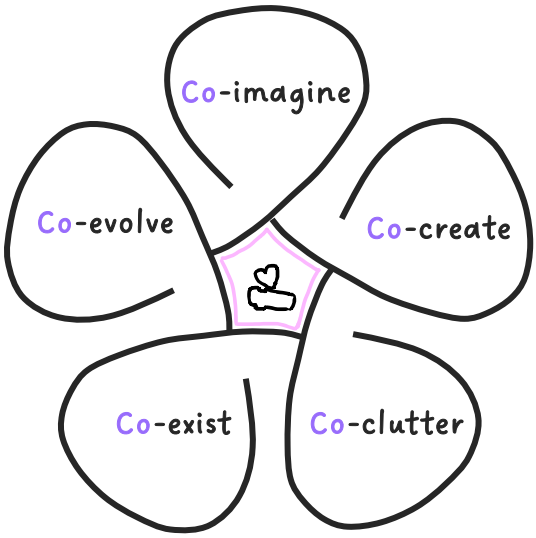


Fig 26: The 5Co's of Co-creative Learning

In any co-creative learning environment, when trust starts to erode, learning and creating stops.

But the wonderful thing is that it's not a static process, we are continually engaged in building and rebuilding our shared context—allowing us to create and recreate trust, openness, authenticity, and of course, with each rebuilding, the increasing capacity to withstand the heat of collisions. Much like the Large Hadron Collider, where particles smash together with each other at near light speed and release new particles, our shared (thermally resistant) context lets us smash together our differing ideas, identities, and communication styles—generating new thoughts, richer insights, and unexpected directions.

It is our differences that make the difference. Co-existing through the flux of co-clutter always leads to co-evolving.

Our ideas have evolved over time through connections with other frameworks that have inspired us—in particular, the Creative Learning Spiral by Mitch Resnick [26], which has long served as a foundation for thinking about how children engage in creative learning. It has informed the design of tools like Scratch and have inspired generations of educators and researchers. It continues to deeply influence how we think about creative learning as an iterative process.

But through our experiences over the past five years in designing, developing, and introducing CoCo to children, we got to see first-hand how in a co-creative learning space, learning and creating does not unfold as an individual process for every learner, it emerges in *between* them. When an environment (physical or digital) is intentionally designed to support active creative collisions between young people's ideas, and projects, and personalities, the stages of co-creative learning naturally appear as an inevitable part of the learning process. We see the 5Co's of Co-creative Learning as an expansion of the spiral where the center of gravity shifts from the learner to their shared context. Children become co-facilitators of one another's learning and creative processes. And an educator's role shifts from helping individual learners' creative processes to cultivating a shared space for them to encounter and experience a rich array of meaningful "co-" experiences with one another.

To further illustrate our thinking about how the ideas of co-creative learning meaningfully build on and extend the creative learning principles, we'd like to share a short playful reflection in the spirit of the wonderful 4Ps:

4P's walk into a co-creative learning space and...

"Projects" discover it's not just about making cool things, but making warm contexts too. It sees children as architects of both their creations and the shared spaces between them. "The thing is being constructed within a context, but maybe constructing the thing is only a context for constructing the context?" Children become not only makers of projects, but also co-makers of the social, emotional, and relational space they inhabit.

"Passion" wobbles at first, then gracefully splits into a trio: "Passion, Dispassion, and Compassion." Passion still hums with that childlike spark and drive. Dispassion brings a calm, Zen-like clarity—children create without being trapped by external validation. Compassion radiates outward—offering care, gratitude, and joy in others' creations. This balanced trio allows young creators to remain fully engaged while staying free from the anxieties of judgement and comparison.

"Peers" peers through the kaleidoscope of co-creation and realizes learning isn't just happening with or from others, but through them. In the space between children, something magical emerges—creating and learning that belongs to neither yet enriches both. Peers are no longer just collaborators or audiences, but become the very medium through which creative imagination flows.

"Play" smiles knowingly, having always understood what the others are just discovering: it was never about metrics or outcomes. Play has always been about the joy of the moment, the process over product, the delight of being fully engaged in Being Creative Together! And play expands beyond tinkering with materials to experimenting with new ways of being in relationship with others.

We hope the 5Co's framework for co-creative learning and the spectrum of ideas it encompasses serve as a helpful lens for others to observe, understand, imagine, and design co-creative learning spaces—where groups of children or adults can experience the power and joy of *being creative together*.

Chapter 17

From Monoculture to CoCo-culture

Monocultures of the mind and screen

While reading an article about monocropping in agriculture, the image in our mind shifted—from vast fields of identical yellow crops to a classroom filled with rows of screens, each displaying nearly identical sprites, code blocks, and animations. The class aimed to teach foundational computing ideas through Scratch—an *endlessly creative* programming language. Yet the teaching approach assigned every student the same tutorial, resulting in nearly identical projects across the room.

Fig 27: Monoculture in classrooms
(image generated from ChatGPT for illustrative purposes)

This kind of educational monocropping is common across subjects. Students are assigned identical tasks, solve the same problems, and are expected to arrive at similar outcomes. These practices simplify planning, instruction, assessment, and classroom management—but at a cost.

Like its agricultural counterpart, educational monocropping can be detrimental to students' motivation, creativity, and relationships.

CoCo-culture: Cultivating Forest Gardens of Co-creativity and Collaboration

Imagine learning spaces for young people…
- where they engage not only with one another's finished work but also with their emerging ideas and unfolding creative processes, in real time?;
- where influence and inspiration are embraced as natural parts of the learning process, rather than being stigmatized as copying or cheating?;
- where collaboration emerges organically from shared interests, rather than being imposed in fixed groups and outcomes;
- where the environment is kept free from comparative metrics that shift focus from exploration to competition?

What kind of learning culture might emerge in such spaces? These questions have guided the development of the **co-creative learning** paradigm underlying CoCo—one that treats learning as happening not just *with* or *from*, but *through* others.

In this chapter, we explore how this paradigm offers possibilities for a new learning culture—a CoCo-culture—which transforms *standardized fields* of learning into *diverse forest gardens* of co-creativity and collaboration. Beyond any tool or platform, CoCo-culture represents a broader mindset and approach to learning that could emerge in any environment designed with these values.

From knowing to designing

Before we dive in, it is important to note that the problem isn't just that monoculture classrooms exist. It's that they persist even when we know better. Even when educators value creativity, collaboration, and learner agency, we continue to design systems that default to uniformity and isolation. Why? Because the challenge isn't knowing—it's designing.

We don't lack learning philosophies. We lack environments and tools that make those philosophies easy to live. And even when some educators manage to embody these values through extra effort, their practices often remain fragile and un-sustained—dependent on personal commitment rather than shared infrastructure. The defaults

don't change. Co-creation stays exceptional instead of expected. It becomes a special project, a one-off workshop, a teacher "going above and beyond." But for a culture to change, what's good must also become what's easy. The values of CoCo-culture—connection, emergence, shared authorship—are widely appreciated in theory. But they're hard to practice unless they are made easy to enact.

The most powerful force in any system is the default.

What if co-creation was the easy choice? This insight reframes what follows. The contrasts between monoculture and CoCo-culture aren't just a matter of preference or pedagogy. They are design choices. They are defaults we can either reinforce or reimagine.

Contrasting Monoculture and CoCo Culture

Table 1: Monoculture vs. CoCo-culture

Attribute	Monoculture classrooms	CoCo-culture learning spaces
Motivations	Need extrinsic motivational fertilizers	Nurture connection as a pathway into learning
Mindsets	Breed competition-oriented mindsets	Cultivate co-creative and collaborative mindsets
Learning process	Limited cross-pollination leading to uniformity	Abundant inspiration through creative collisions
Type of collaboration	Prescribed collaboration	Inclusive of multiple ways of being creative together
Identities	Individualized identities	Co-constructing new shared identities
Assessment	Focus on assessing standardized outcomes	Emphasizes reflections on co-creative and collaborative process
Educator role	Instructor and manager	Facilitator of co-creative learning process

Rethinking AI in Education: Monoculture or CoCo-culture

If we look closely, most learning environments—online and offline—are built with monocultural assumptions, designed around isolation masquerading as personalization. AI tutors get increasingly adept at adapting to individual learners, yet remain unaware of the collective learning unfolding around them. Platforms track individual progress but rarely attend to the intelligence emerging between students. Assessment systems measure what is learned and created individually but often overlook what can be discovered and created—together. We've become so sophisticated at personalizing learning that we've forgotten learning is fundamentally a social, cultural, and co-creative act.

The metaphor of Monoculture versus CoCo-culture offers a lens for thinking about AI in education. Will AI be used to create hyper-personalized, hyper-isolated learning experiences, where students interact primarily with machines? Or will we use it to design for connection—surfacing relationships between ideas, supporting co-creation, and making shared learning visible?

CoCo envisions the latter. Used wisely, AI can amplify the "co" in co-creation—not just the "c0" of copilots and companion bots. Let's not confuse c0-creating (with machines) with co-creating (with humans). The root "co" stands for our human wholeness—rooted in emotion, shared experience, and meaning. It is through this "co" that we build shared futures.

Chapter 18

Bringing CoCo (and CoCo-culture) to communities around the world

Fig 28: Interest for CoCo from around the world

At the time of writing this dissertation, thousands of educators, parents, and young people from **85+ countries** have signed up to bring CoCo into their classrooms and communities—sharing their excitement for the new possibilities it opens up for collaborative creativity, as well as the underlying ideas and values.

> "You've made a platform that's **human in ways that few platforms even think to consider**."

> "I have honestly not stopped thinking about its **transformative potential to fundamentally change teaching and learning**, and not just in CS classrooms."

"Thrilled about how it's based on **real-time co-presence**! I love that the focus is on a shared creative space, not necessarily explicit collaboration."

"This opens a **whole new way to collaborate** on creative computing projects!"

"Wow! **Love the idea of a "self-less social platform"** for kids."

"I like that CoCo makes learning **collaborative, fun, and human**, and that the environment is intentionally built to **de-emphasize comparison** with others: no profiles, followers, or likes."

"A **gorgeous, creative, and humane online space** for kids!"

This work has been rooted in intuitions, intentions, and values that we deeply care about at a personal level. And it has been wonderful to see how these ideas and values have resonated with others. It makes us feel energized and committed to keep working toward bringing CoCo in the hands of young people everywhere—with the hope that it can play a meaningful part toward helping them experience the CoCo-culture.

"

> If we create, inside of ourselves, a respect for beauty, then we will create beauty outside of ourselves…We might find that what makes us feel peaceful makes others feel peaceful, that what makes us feel sublime makes others feel sublime, that what makes us feel alive makes others feel alive. We might find that what fills us with wonder and hope fills others with wonder and hope. We might find that what is beautiful to us is beautiful to others.
>
> — *Syntax and Sage, Sep Kamvar [27]*

Chapter 19

Extending CoCo | Little Language Models (LLMs): Empowering children to be future AI modelers

Little Language Models [28] is a microworld that we have been building within CoCo to support young people in exploring powerful ideas around Artificial Intelligence. This chapter dives into the details of this work[3]. While the chapter is not directly related to the core themes of this dissertation, we highlight this work to share how we are continuing to extend CoCo in new ways while keeping the underlying co-creative approach at the center of children's learning experiences.

We want to start by stating that…

Future will NOT be shaped by AI; it will be shaped by the modelers of AI.

If children are not empowered to develop creative fluency with the foundational ideas underlying AI, we risk raising a generation that—instead of modeling AI systems in the future—might end up being modeled by them.

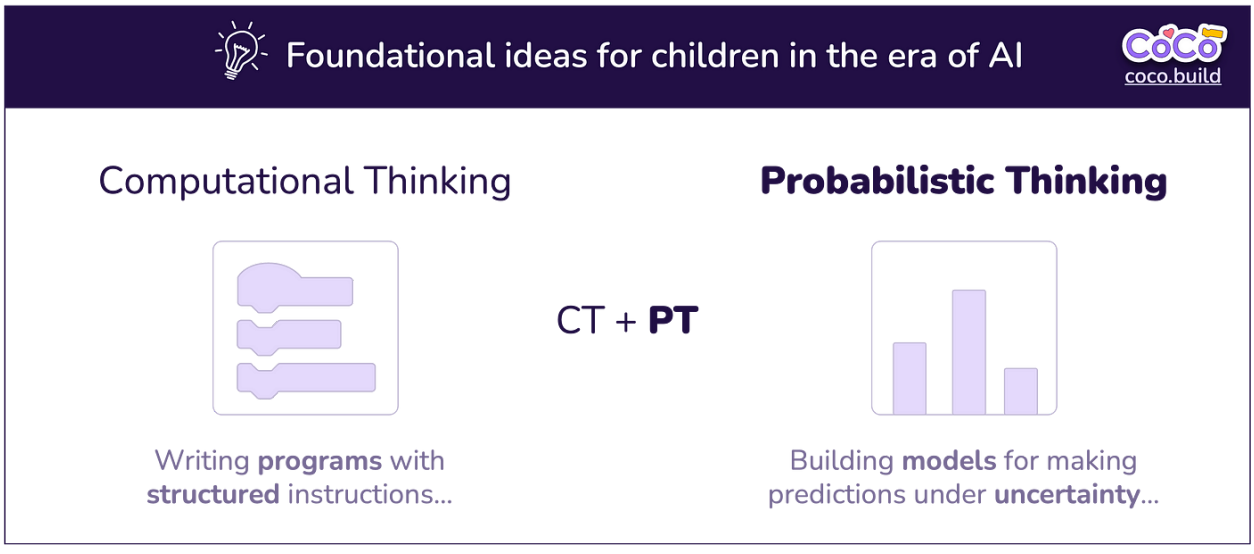
[3] This chapter has been adapted from our original blog post on Medium [29].

We believe it's essential for young people today to not just learn how to 'use' the new *magical* AI tools. They should also have access to tools that help make AI *unmagical* for them—by letting them explore and engage early on with the powerful ideas that underlie these powerful technologies.

💡 Introducing "Probabilistic Thinking" as a new foundational fluency for children in the era of AI

In any era, the most important ideas for children to explore are often the ones that are not narrowly restricted to a particular concept or skill, but instead are foundational to children's understanding of the world. They help them develop new ways of thinking about their own thinking and learning, widely expanding the space of possibilities they can imagine.

In the digital era, *Computational Thinking (CT)* emerged to be one such idea and is now recognized as an essential literacy for learners globally. As we enter the AI era, it is both a critical and opportune time to reassess what types of ideas and fluencies will be most important for young people growing up today. Through this work, we aim to bring attention to a *timeless* foundational idea that has become especially *timely* now—the idea of *Probabilistic Thinking*.

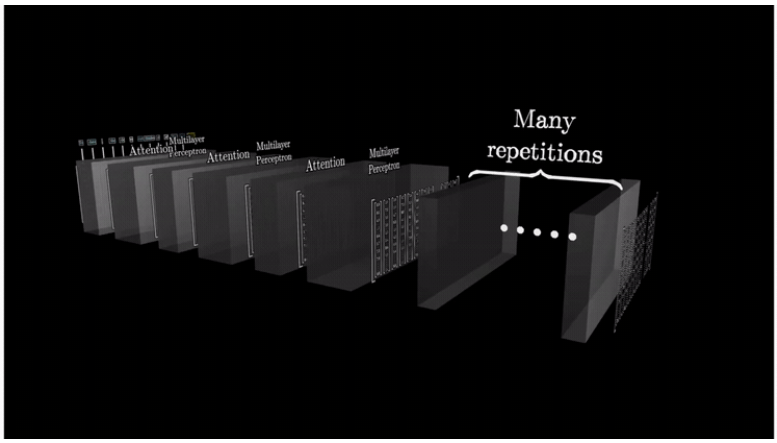


CT + PT as foundational ideas for children in the era of AI

Probabilistic Thinking (PT) can be described as a decision-making approach that involves making predictions under uncertainty by building and using models that learn from data. It provides a powerful framework to understand intelligence and learning processes, both artificial and human.

Just as **CT** underpins computer *programming*, the act of writing structured instructions for computers; **PT** is fundamental to probabilistic *modeling*—the process of building mathematical models that can quantify uncertainties and make predictions by learning from data.

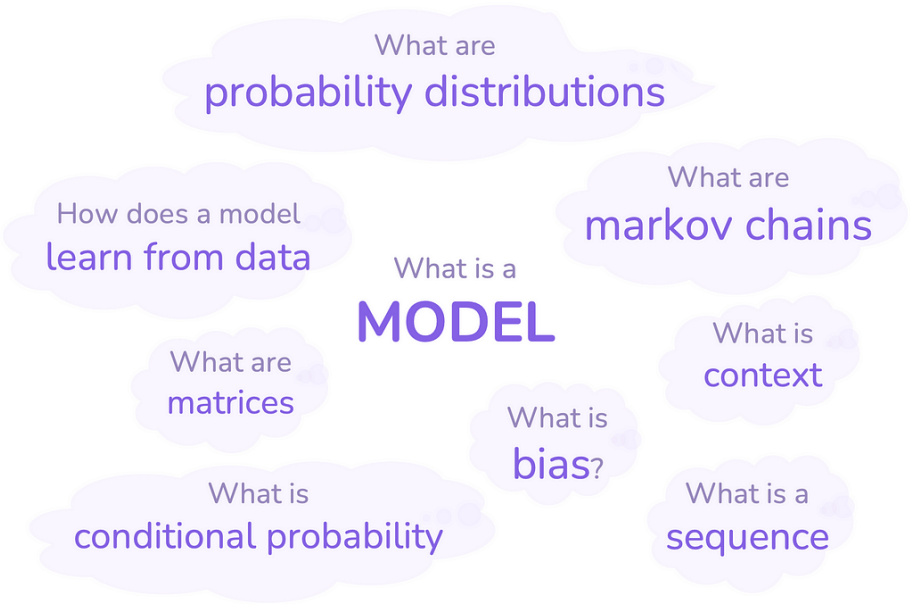
The ideas of Probabilistic Thinking, Modeling, and Learning lie at the core of how Generative AI and Large Language Models (LLMs) like ChatGPT work.



Gif generated from this video by 3Blue1Brown explaining GPT

For instance, when you input a text prompt into ChatGPT, the model is able to generate a response by using probabilities (derived from processing training data) to predict the most likely next word or phrase. This ability makes LLMs incredibly useful and powerful across a wide variety of tasks, including for generating code. And so while the core ideas of computational thinking will continue to hold value, it is evident that the future builders and computer science researchers, will not just be writing *programs* but increasingly engage in tuning, tinkering, and building *models*. How can we empower children to bring this (now ubiquitous) word *'model'* into their own worlds and imaginations so it becomes a part of their thinking and vocabulary?

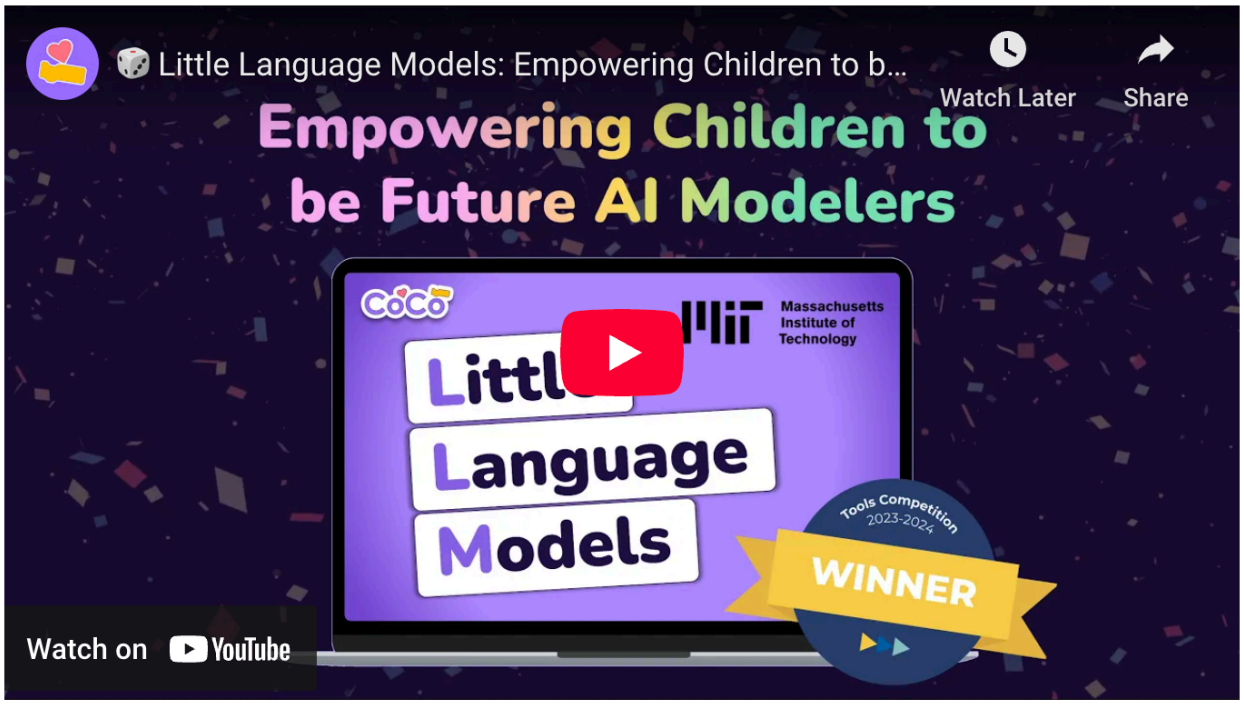
How can we support educators to make the foundational ideas of probabilistic thinking, modeling, and learning accessible and engaging for young people?



Meet 'Little Language Models'—🎲🔤🎨🎸

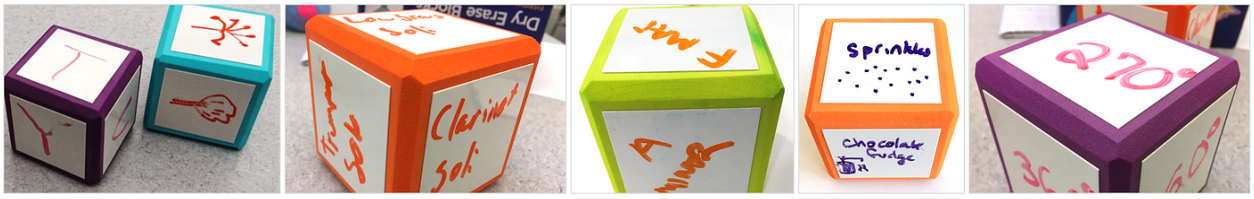
'Little Language Models' (*Little Models* for short) is a co-creative mathematical microworld for children (ages 8–16) within our CoCo platform (coco.build). It is especially designed to be developmentally appropriate for young learners, helping them explore sophisticated probabilistic ideas and concepts underlying AI technologies—in ways that are co-creative, playful, and personally meaningful for them.
The tool offers custom probabilistic blocks and manipulatives that let children *build their own little models using their own little data*. The data can be in any form, including text, images, sounds, etc. Children can dynamically tinker with their models and use them to collaboratively explore new kinds of creative and generative projects with their peers. For instance, they can make models for generative art, stories, or music; build adaptive multiplayer games, or interactive visualizations based on real-time data from their peers, or even advanced projects using Markov blocks such as teaching a computer to learn how to draw, among an endless variety of other possibilities.

In the spirit of *exploration* before *explanation*—here's a short 1-min video that gives a glimpse into the tool. Make sure to turn the sound ON to listen to our special *little models* song 🎙…Disclaimer: The lyrics and voce are **not** AI generated.



A little glimpse into the Little Models microworld: https://youtu.be/zDHdKREA5og

🎲 DICE: A playful, mathematically-rich object to make the abstract idea of a "model" concrete for children



*Ideas become **powerful** only when they become **personal**.*

In conceiving and designing *Little Models*, one of our key insights early on was to ground the abstract idea of a 'model' in a mathematically-rich object that is familiar, concrete, and playful for children: the *dice*. But unlike standard dice that have fixed numbers from 1 to 6 on the sides, the dice in the *Little Models* microworld are designed to be infinitely customizable. The metaphor of physical dice makes it easier for young people to start imagining the little models (or digital dice) they'd like to build in the

environment. We've often been delighted by the diversity of ideas children come up with when they discover that they can use the tool to **make a model for anything they like**—*names, musical notes, or even ice cream flavors!*

Children can create their own digital customizable dice in the environment.

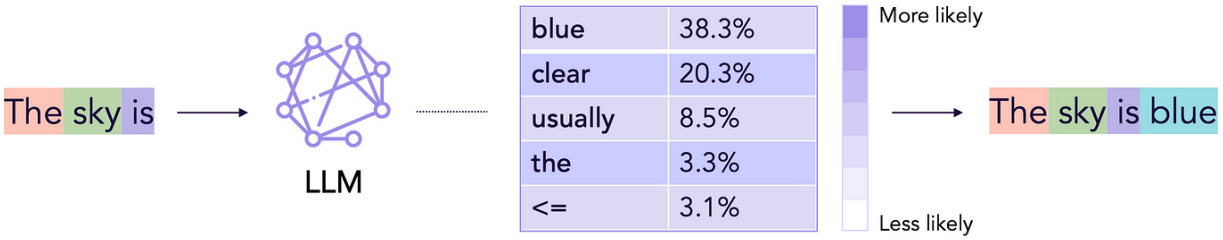
🔍 Building their own "Little" Language Models can help demystify "Large" Language Models for children

In our workshops with young people, we typically follow a two-part structure. The first part involves hands-on explorations with the tool. During this time, children spend time imagining, building, and tinkering with their models to co-create a wide variety of generative projects with their peers in the real-time computational environment of CoCo. In the second part, we engage in a reflective discussion with the group about the key ideas and concepts they explored in the process.

Below, we highlight some of the conceptual, mathematical, algorithmic, and ethical lenses for understanding Generative AI models that we have seen children discover and develop through their creative explorations within the *Little Models* microworld:

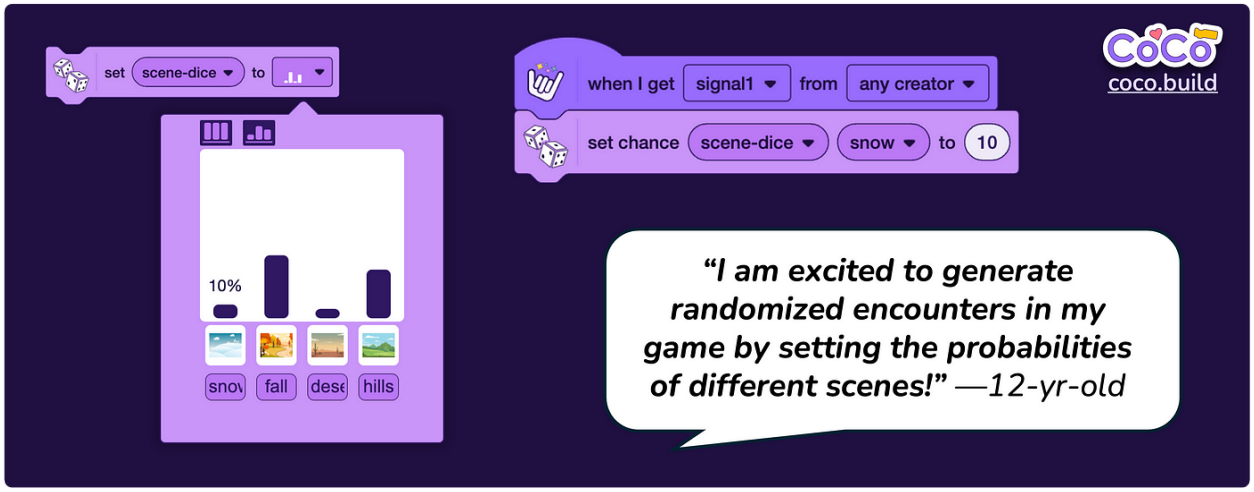
1) 📶 Behind the magic, there's a "Probability Distribution"

"I now know more about how a model works."—13-year-old



Remix of an image from an NVIDIA developer blog about LLMs

The most salient shift we see in students before and after our workshops is in their understanding of the powerful idea of *probability*. Existing approaches to introducing probability in classrooms often reduce it to mere calculation of fractions pertaining to narrow and impersonal events such as coin tosses, lotteries, or spinners. Even when doing workshops with high schoolers, we find that most of them often think of probability only in terms of specific formulaic descriptions. It's not often intuitive to students how it is one of the key ideas underlying Generative AI systems.



Children can dynamically update the probabilities in their models.

In the *Little Models* microworld, children are not only able to create their own dice (or models), but also dynamically update the underlying probability distribution and visualize the effect in real-time in the kind of samples their model generates. They start to see *probability* as a playful clay like material that they can easily tinker with to explore a wide variety of creative possibilities. In the process, they are able to *generalize* the underlying mathematical ideas and see how they can be applied across different

contexts.

2) 🎨 It's not just text, AI can be "Multimodal"

"I used to think of AI as solely chatGPT, but now I see it can have many more capabilities."—14-year-old

Children can make little models using multiple types of data modalities

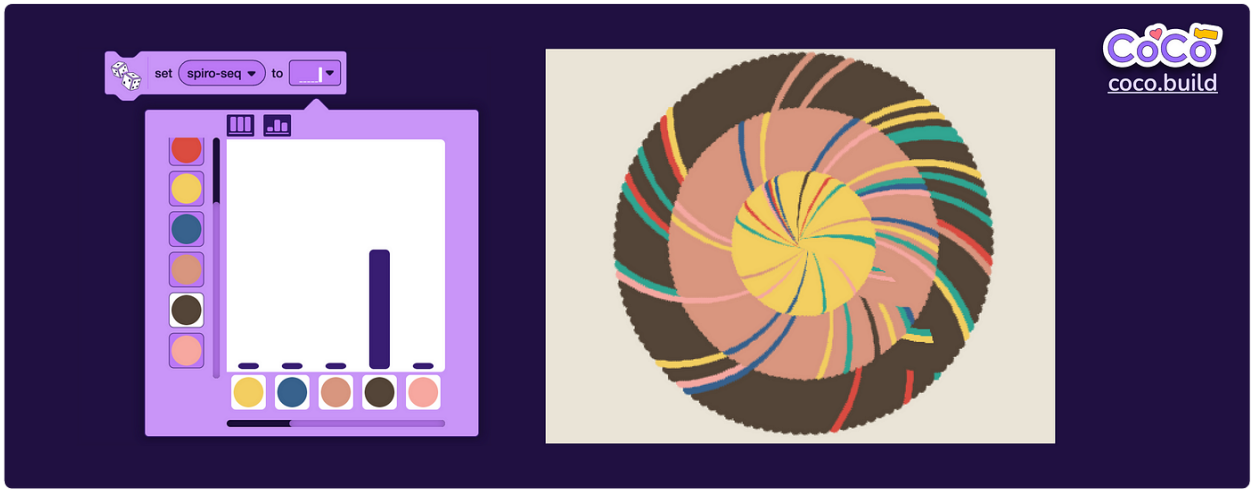
The *Little Models* tool is not limited to supporting only text data. It is designed to support multiple modalities including text, images, sounds, etc. Students are thus able to build generative models using music, text, colors, shapes, images, and also micro:bit sensor values as data. This helps expand their understanding of what constitutes a "language" beyond natural (text-based) languages. They start to see how their expressions made out of musical notes, drawing strokes, or even structures made from LEGO blocks can all be viewed as "languages," each with its own elements and rules for composition. By recognizing these parallels, students are able to better grasp how modern Generative AI systems are able to produce not just coherent text, but also generate images, compose music, or create 3D designs, all by learning the patterns of the underlying "language."

3) 📊 A little bit of "Training" goes a long way

"I had always wondered how things like chatgpt worked."—13-year-old

The *Little Models* microworld is intentionally designed to support learners across ages by providing a scaffolded experience for beginners to get started easily while also enabling sophisticated possibilities for more advanced learners. After creating and playing with simple weighted models, students can progress to building more complex models like Markov models. The intuitive interface allows learners to immediately see how the sequence of elements in their input data "trains" their Markov model and updates its underlying probabilities. This helps them draw parallels with key concepts in Generative AI as mentioned below:

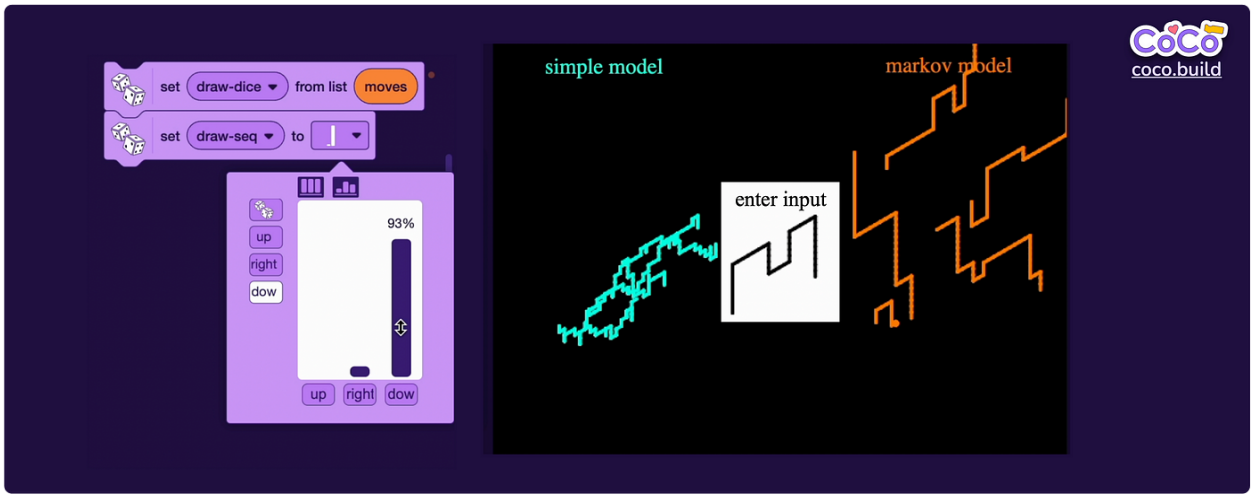
1. Concept of 'Sequence': Children are able to make connections with how Generative AI systems also process and generate sequences of tokens one at a time, forming cohesive outputs such as essays, images, or music. For instance, using the tool, students can enter a sequence of notes and train their model to generate new musical sequences, or enter a sequence of colors to generate new patterns based on that.



Generating patterns based on input sequence of colors

2. Concept of 'Context Length': A defining feature of systems like ChatGPT is their ability to maintain context during a conversation, providing answers by considering a large amount of information (words / tokens) from previous interactions. In contrast, the simple Markov models children create typically have a context length of 1, meaning the model generates the next token based only on the last one. Students are able to explore how even a context length of 1 significantly improves their model's output compared to a simple weighted model with a context length of 0. The same can be

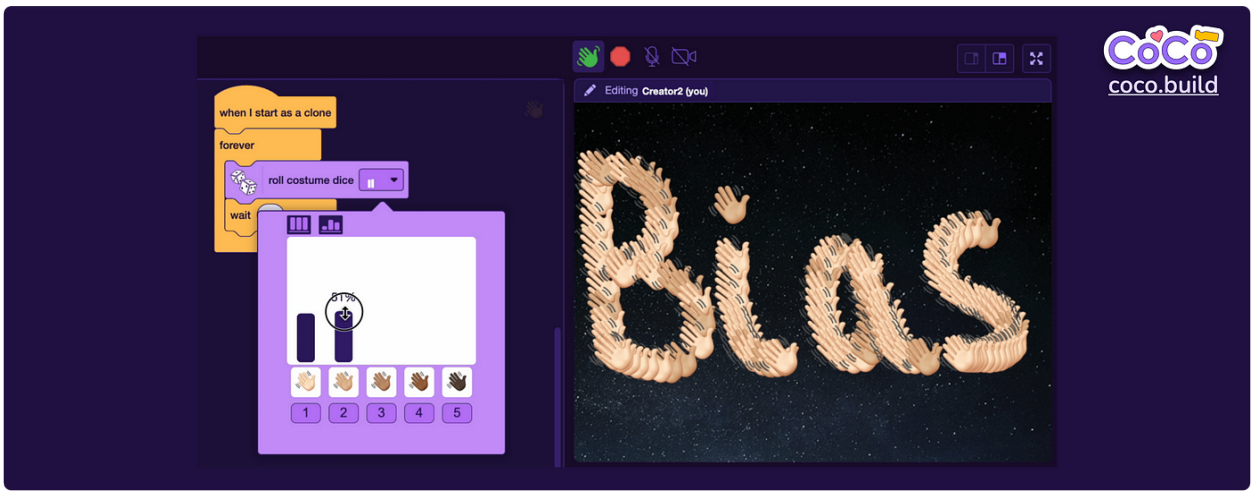
seen in this generative drawing model made by a student by entering a sequence of strokes as training data.



Training a Markov model based on input strokes

4) 🟨 "Bias" in output stems from input

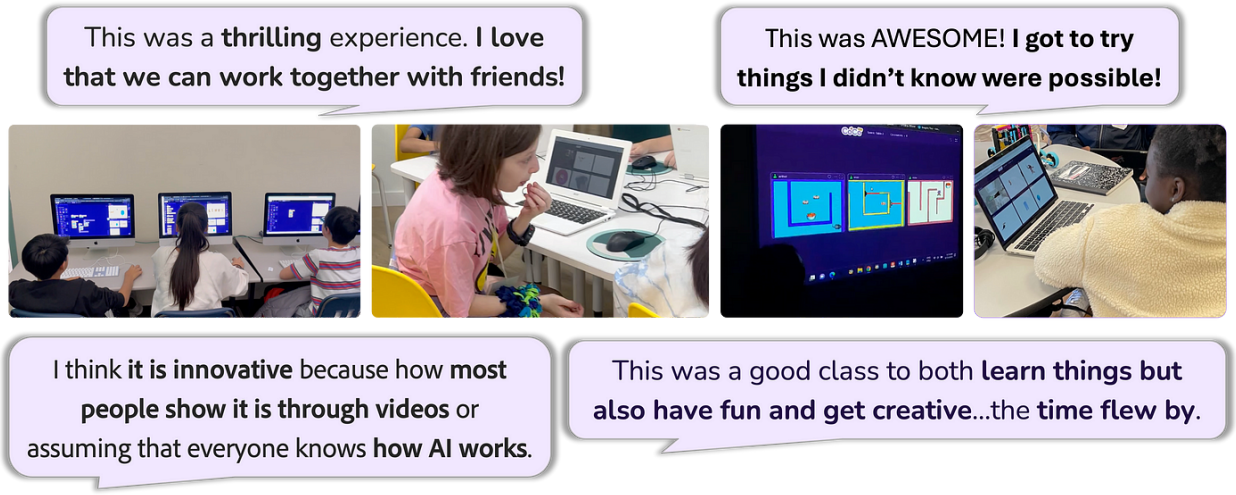
"I have a better sense of what decisions are being made behind the scenes and how that can be mimicked on a smaller scale."—15-year-old



Visualizing 'bias' in a model

One of the critical ideas we've seen students getting to *experience* for themselves is around how *bias* can creep into AI models. The tool makes it concrete and visible to children that the model will draw out samples according to the distribution that they have set. That is, if they have more of yellow in the input data, the chance of rolling a

yellow will be high. This provides a rich and concrete context for educators to have conversations with children about issues of bias and fairness in AI models.



Pilots with students

Through children's feedback in the pilot workshops and our interactions with them, we've seen first-hand how building their own little models with the tool can help them develop interests, intuitions, and insights about how Large Language Models like ChatGPT and others operate. The abstract ideas and terminologies that are used in explaining the workings of AI systems often remain too obscure and inaccessible for young learners (and even adults!). We hope the *Little Models* microworld can serve as a rich playground for educators to support children in developing a language and lens for these big ideas—helping them see AI as *manipulable,* not magical.

💭 Envisioning a new future for AI education: Toward "Intelligetics"

In this final section, we'd like to zoom out and *scribble* some of our thoughts on what the future of AI education can look like. We imagine that we will soon see an

emergence of a new, inclusive, and interdisciplinary field in academia, which will then also be introduced as a core subject in K-12 education, much like math or social science. We propose calling it '*Intelligetics*' (🗣️ Intelli-ge-tics).

Intelligetics will focus on the comprehensive study of intelligence—integrating human intelligence studies with the development and analysis of artificial intelligence systems. It will synthesize powerful ideas from a diverse array of existing disciplines, including: computer science, cognitive science, neuroscience, mathematics, artificial intelligence, philosophy, linguistics, social science, and learning sciences. It will attract people from varied backgrounds with diverse interests. Practitioners in the field—let's call them '*Intelligicians*' (🗣️ Intelli-gi-cians)—will bridge multiple areas of expertise, developing holistic approaches for advancing our understanding and capabilities in both natural and artificial intelligence, as well as their interplays.

Teaching children to be "Intelligicians"

How will we nurture the next generation to be *Intelligicians*? In 1971, Seymour Papert published a memo at the A.I. Laboratory at MIT titled—*Teaching children to be mathematicians vs. Teaching children about mathematics.* He argued that "Children will become more proficient in mathematics if they *do* mathematics rather than merely learn *about* mathematics." We deeply resonate with these words and the underlying constructionist paradigms of teaching and learning they represent.

We see our ideas and work on *Little Language Models* as a little but important first step toward offering children a rich, co-creative microworld where they can actively engage in *doing* AI (or *Intelligetics*) with their peers, rather than merely learning *about* AI passively. As we look ahead, we are excited to continue imagining and building new microworlds that help empower children everywhere to see themselves as future AI builders, scientists, and *Intelligicians*—who have the skills and will to shape the future in ways that best serve humanity.

Fig 29: Children as 'Little Modelers' (image from ChatGPT)

Chapter 20

Looking ahead:
Centering the timeless in timely

Every era brings its own urgencies, its own "timely" opportunities and concerns. But as we collectively look to the future, design for the future, explore the exciting new possibilities that new technologies offer us, and think about what *is* going to change in the era of AI—can we center what *is not* going to change, regardless of any era?

Our enduring needs: Joy. Connection. Expression. Play. Togetherness. Calm. Harmony. And everything else that resembles these. These experiences and values bring us closer to who we are, and we cherish them across generations and geographies. They do not change with the changing technologies. But they do need conscious cultivation and protection.

They need *centering*.

What is centering?

In architectural terms, "centering" refers to a temporary support structure—framing that holds up arches and vaults during construction until they can bear their own weight. Once the structure hardens and finds its strength, the centering is removed, leaving behind something that can stand for centuries.

This dissertation has been, in many ways, an exercise in centering—providing support for ideas that we hope will one day stand on their own without needing any justification. **We actually wish for a future where no one else has to write a dissertation about designing for human connection, co-creativity, and calm.**

Because in that future, these will not be add-ons or good-to-haves—but rather assumed—foundational, built-in—in every creative tool, every learning space, every social platform.

A future where the dominant individual-centric paradigms of today will be outdated. Where being creative will mean being creative *together*—in myriad ways.

A future where the technologies and environments we engage with will make us feel more *present*, more calm, more in tune with ourselves, more in harmony with others—not anxious, disengaged, and disconnected.

A future where the mediums we design serve as microworlds to continually construct, deconstruct, and reconstruct our self-identities—evolving the self into the self-*less*.

A future where to co- will not be against the norm, but the *default*.

In this spirit, we hope the ideas, technologies, possibilities, and frameworks we have shared through this work serve as *centering*—that can help support and shape *everlasting* structures of such futures.

We co-, therefore we are.

The most *timely* work is always in service of the *timeless*.

To co- is *timeless*.

Afterword(s)

Through the years of shared explorations and countless conversations, we have co-evolved a vocabulary—a set of words, framings, and metaphors—that have helped us imagine new possibilities and articulate new ways of thinking, seeing, doing, and being. We hope they help spark new imaginations, possibilities, and lenses for others who encounter them.

To co- is human.
To cO- AI.

Constructing **things,**
Constructing **context**

Knobs
over
Switches

Being. Creative. Together.

Co-creative Learning
creating and learning
THROUGH one another

CoCo (coco.build)
Connection and Construction

From **Monoculture**
to **CoCo-culture**

The 5Co's
Co-imagine
Co-create
Co-clutter
Co-exist
Co-evolve

Passion,
Dispassion,
Compassion

Head, hands, heart,
and **hüm**

Designing
"self-less" social systems

Children as facilitators
of each other's learning

Centering the
timeless in timely

From
future to present

Creating the conditions
for **collision**

May we continue to **co-evolve**
through one another...

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This little bouquet
has the names of
all the people and
all the communities
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and will forever be.

I hope each of you
can spot your name.

To co- is to be **with**

thank you for
being **with** me

in my heart, in my mind,
and in the spaces between.

in who I am, in who I will be,
and in the me between.

- shruti